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CS229 Machine Learning

Probability Distributions

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Binary Variables

- *Bernoulli distribution*: single binary random variable $x \in \{0,1\}$
- Example: flipping (fair or unfair) coin

$$p(x = 1|\mu) = \mu \qquad p(x = 0|\mu) = 1 - \mu \qquad \mu \in [0,1]$$

$$Bern(x|\mu) = \mu^x \, (1 - \mu)^{1-x}$$

- Expectation and variance

$$\mathbb{E}[x] = 1.\mu + 0.(1 - \mu)$$

$$var(x) = \mathbb{E}[x^2] - \mathbb{E}[x]^2 = \{\mu + 0.(1 - \mu)\} - \mu^2 = \mu(1 - \mu)$$

Binary Variables

- $\mathcal{D} = \{x_1, \dots, x_N\}$ of observed values

$$p(\mathcal{D}|\mu) = \prod_{n=1}^{N} p(x_n) = \prod_{n=1}^{N} \mu^{x_n}(1-\mu)^{1-x_n}$$

- Estimating μ with MLE

$$\ln p(\mathcal{D}|\mu) = \sum_{n=1}^{N} (x_n \ln \mu + (1 - x_n) \ln(1-\mu))$$

Binary Variables

$$S = \sum_{n=1}^{N} x_n$$

$$\ln p(\mathcal{D}|\mu) = S \ln \mu + (N - S) \ln(1 - \mu)$$

$$\frac{d}{d\mu} \ln p(\mathcal{D}|\mu) = \frac{S}{\mu} - \frac{N - S}{1 - \mu} = 0$$

$$S(1 - \mu) = (N - S)\mu \Rightarrow S = \mu(S + N - S)$$

$$\mu_{ML} = \frac{S}{N} = \frac{1}{N} \sum_{n=1}^{N} x_n$$

Binary Variables

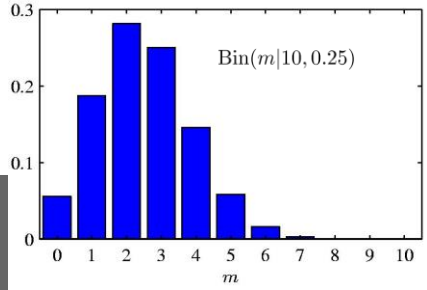
- *Binomial distribution*: represents the number m *of successes* in N independent Bernoulli trials, each with success probability μ

$$Bin(m|N,\mu) = \binom{N}{m} \mu^m \, (1-\mu)^{N-m}$$

$$\mathbb{E}[m] = \sum_{m=0}^{N} m Bin(m|N,\mu) = \sum_{m=0}^{N} m \binom{N}{m} \mu^m \, (1-\mu)^{N-m}$$

$$m\binom{N}{m} = m\frac{N!}{(N-m)!\,m!} = m\frac{N(N-1)!}{m\big(N-1-(m-1)\big)!\,(m-1)!}$$

$$= N\binom{N-1}{m-1}$$



Binary Variables

- *Binomial distribution:*

$$\mathbb{E}[m] = N \sum_{m=1}^{N} \binom{N-1}{m-1} \mu^m \, (1-\mu)^{N-m}$$

$$= N \sum_{m=0}^{N-1} \binom{N-1}{m} \mu^{m+1} \, (1-\mu)^{N-(m+1)}$$

$$= N\mu \sum_{m=0}^{N-1} \binom{N-1}{m} \mu^m \, (1-\mu)^{N-1-m} = N\mu(\mu + (1-\mu))^{N-1} = N\mu$$

Binary Variables

- *Binomial distribution:*

$$\mathbb{E}[m^2] = \sum_{m=0}^{N} m^2 \binom{N}{m} \mu^m \, (1-\mu)^{N-m} = \sum_{m=0}^{N} [m(m-1)+m] \binom{N}{m} \mu^m \, (1-\mu)^{N-m}$$

$$= N(N-1) \sum_{m=2}^{N-2} \binom{N-2}{m-2} \mu^m \, (1-\mu)^{N-m} + N\mu$$

$$= N(N-1)\mu^2 \sum_{m=0}^{N-2} \binom{N-2}{m} \mu^m \, (1-\mu)^{N-2+m} + N\mu = N(N-1)\mu^2 + N\mu$$

$$var(m) = \mathbb{E}[m^2] - \mathbb{E}[m]^2 = N^2\mu^2 - N\mu^2 + N\mu - N^2\mu^2 = N\mu(1-\mu)$$

Binary Variables

- *Beta distribution*: prior distribution for developing a Bayesian treatment for binary random variables
- *Conjugacy* means that prior and likelihood function have the same functional form, then the posterior distribution will also have the same functional form

$$Beta(\mu|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\mu^{a-1}(1-\mu)^{b-1}$$

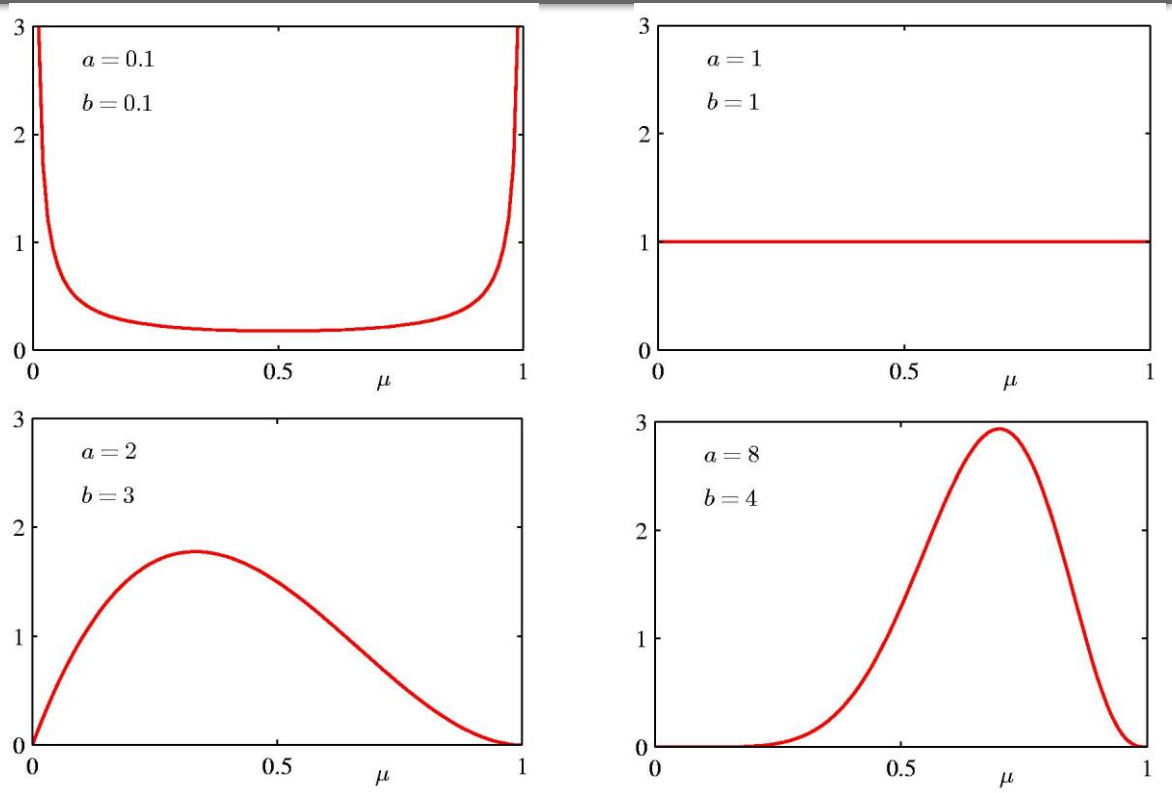
$$\Gamma(x) \equiv \int_0^\infty u^{x-1}e^{-u}du \qquad \int_0^1 Beta(\mu|a,b) = 1$$

Binary Variables

- *Beta distribution*: a, b are hyperparameters

$$\mathbb{E}[\mu] = \frac{a}{a+b}$$

$$var(\mu) = \frac{ab}{(a+b)^2(a+b+1)}$$



Binary Variables

- *Bayesian Bernoulli:*

$$p(\mu|m,l,a,b) = \prod_{n=1}^{m+l} \mu^{x_n}(1-\mu)^{1-x_n}\, Beta(\mu|a,b)$$

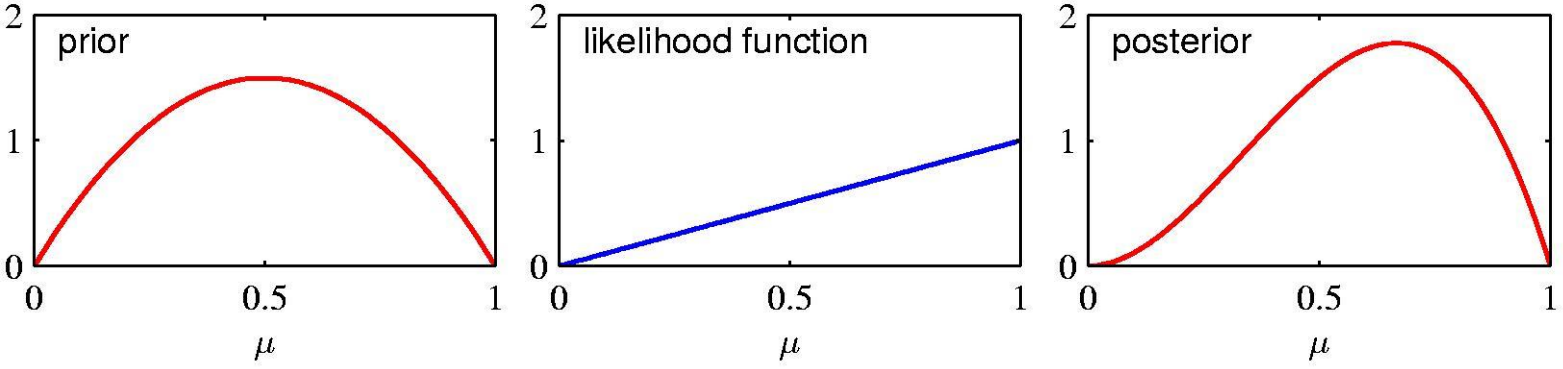
$$= \frac{\Gamma(m+a+l+b)}{\Gamma(m+a)\Gamma(l+b)} \mu^{m+a-1}(1-\mu)^{l+b-1}$$

$$= Beta(\mu|a+m, b+l)$$

- Hyperparameters a, b can be interpreted as number of prior observations

Binary Variables

- *Posterior distribution: sequential approach to learning*



$$a_N = a + m \quad b_N = b + l$$

- Allows for considering historical data as a prior

Binary Variables

- *Posterior distribution:* as the size of the data set N increase $a_N \to m \quad b_N \to l$

$$\mathbb{E}[\mu] = \frac{a_N}{a_N + b_N} \to \frac{m}{m + l} = \mu_{ML}$$

$$var(\mu) = \frac{a_N b_N}{(a_N + b_N)^2 (a_N + b_N + 1)} \to 0$$

Binary Variables

- Prediction under the Posterior: What is the probability that the next coin toss will land heads up?

$$p(x = 1, a, b, \mathcal{D}) = \int_0^1 p(x = 1|\mu) p(\mu|a, b, \mathcal{D}) d\mu$$

$$= \int_0^1 \mu p(\mu|a, b, \mathcal{D}) d\mu = \mathbb{E}[\mu|a, b, \mathcal{D}] = \frac{a + m}{a + m + b + l}$$

Multinomial Variables

- Generalization of Bernoulli distribution to more than two outcomes
- Realized through 1-of-K encoding $(K = 6)$

$$\boldsymbol{x} = (0,0,1,0,0,0)^T$$

$$\sum_{k=1}^{K} x_k = 1 \qquad p(x_k = 1) = \mu_k \qquad p(\boldsymbol{x}|\boldsymbol{\mu}) = \prod_{k=1}^{K} {\mu_k}^{x_k}$$

$$\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_K)^T \quad \sum_{\boldsymbol{x}} p(\boldsymbol{x}|\boldsymbol{\mu}) = \sum_{k=1}^{K} \mu_k = 1$$

$$\mathbb{E}[\boldsymbol{x}|\boldsymbol{\mu}] = \sum_{\boldsymbol{x}} \boldsymbol{x} p(\boldsymbol{x}|\boldsymbol{\mu}) = (\mu_1, \mu_2, \dots, \mu_K)^T = \boldsymbol{\mu}$$

Multinomial Variables

- Dataset $\mathcal{D}$ of N independent observations

$$p(\mathcal{D}|\boldsymbol{\mu}) = \prod_{n=1}^{N}\prod_{k=1}^{K}\mu_k{}^{x_k} = \prod_{k=1}^{K}\mu_k{}^{\sum_n x_{nk}} = \prod_{k=1}^{K}\mu_k{}^{m_k}$$

- Finding $\boldsymbol{\mu}$ using MLE

$$\ln p(\mathcal{D}|\boldsymbol{\mu}) = \sum_{k=1}^{K} m_k \ln \mu_k + \lambda\left(\sum_{k=1}^{K}\mu_k - 1\right)$$

Multinomial Variables

- Dataset $\mathcal{D}$ of N independent observations

$$\frac{\partial \ln p(\mathcal{D}|\boldsymbol{\mu})}{\partial \mu_k} = \frac{m_k}{\mu_k} + \lambda = 0 \Rightarrow \mu_k = -\frac{m_k}{\lambda}$$

- Solving for Lagrange multiplier λ

$$1 = \sum_{k=1}^{K} \mu_k = -\frac{1}{\lambda} \sum_{k=1}^{K} m_k \Rightarrow \lambda = -N \Rightarrow \mu_{k\,ML} = \frac{m_k}{N}$$

Multinomial Variables

- Multinomial distribution:

$$Mult(m_1, m_2, \dots, m_K | \boldsymbol{\mu}, N)$$

$$= \binom{N}{m_1, m_2, \dots, m_K} \prod_{k=1}^{K} {\mu_k}^{m_k}$$

$$\mathbb{E}[m_k] = N\mu_k$$

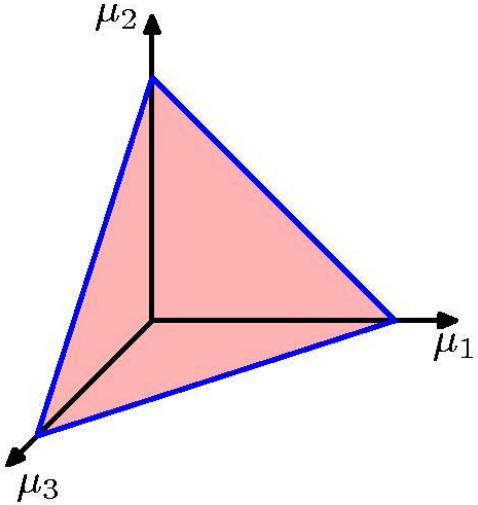
$$var(m_k) = N\mu_k(1 - \mu_k)$$

Multinomial Variables

- Dirichlet distribution:

$$Dir(\boldsymbol{\mu}|\boldsymbol{\alpha}) = \frac{\Gamma(\sum_{k=1}^{K} \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_K)} \prod_{k=1}^{K} \mu^{\alpha_k - 1}$$

- Distribution probabilities confined in a $K-1$ simplex forming barycentric coordinates due to $\sum_{k=1}^{K} \mu_k = 1$



Multinomial Variables

- Posterior distribution using Dirichlet prior:

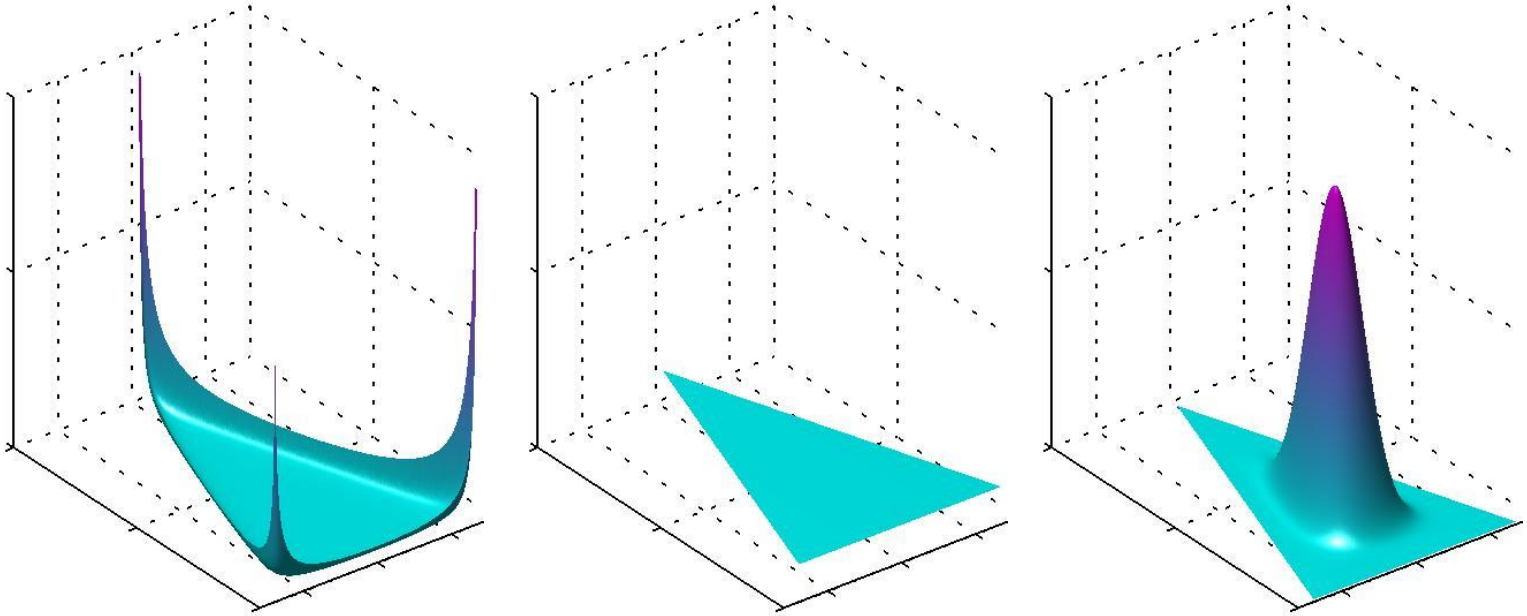
$$p(\boldsymbol{\mu}|\mathcal{D},\boldsymbol{\alpha}) \propto p(\mathcal{D}|\boldsymbol{\mu})\, p(\boldsymbol{\mu}|\boldsymbol{\alpha}) \propto \prod_{k=1}^{K} \mu_k^{\alpha_k+m_k-1}$$

$$p(\boldsymbol{\mu}|\mathcal{D},\boldsymbol{\alpha}) = Dir(\boldsymbol{\mu}|\boldsymbol{\alpha}+\boldsymbol{m})$$

$$= \frac{\Gamma(\sum_{k=1}^{K}\alpha_k + N)}{\Gamma(\alpha_1+m_1)\ldots\Gamma(\alpha_K+m_K)} \prod_{k=1}^{K} \mu_k^{\alpha_k+m_k-1}$$

Multinomial Variables

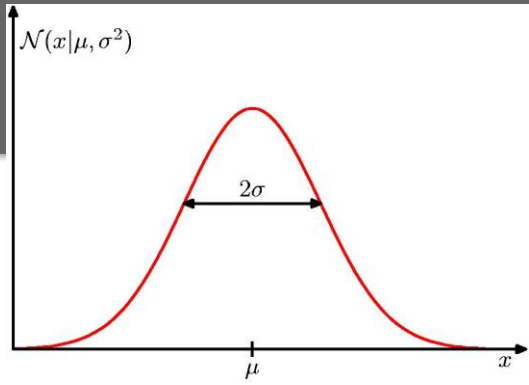
- Dirichlet distribution plot:



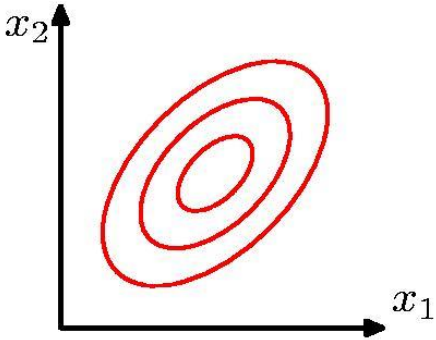
$\alpha_k = 10^{-1}$ $\alpha_k = 10^{0}$ $\alpha_k = 10^{1}$

Gaussian Distribution

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

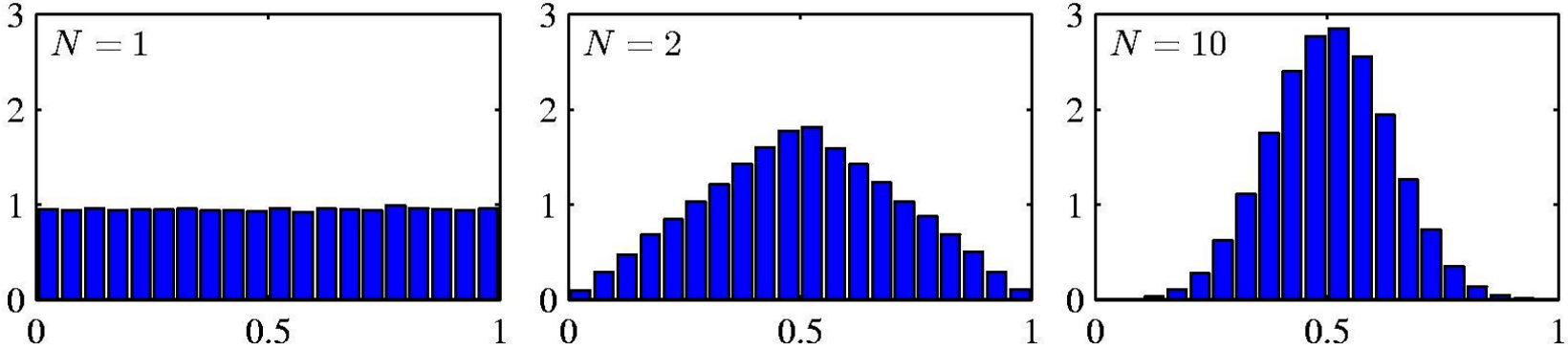


$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}}} \frac{1}{|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}-\boldsymbol{\mu})\right\}$$



Gaussian Distribution

- *Central Limit Theorem:* The distribution of the sum of N i.i.d. random variables becomes increasingly Gaussian as N grows
- Example: N uniform $[0,1]$ random variables



Gaussian Distribution

- Geometry of Multivariate Gaussian: Mahalanobis distance from $\boldsymbol{\mu}$ to $\boldsymbol{x}$ expressed in quadratic form

$$\Delta^2 = (\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})$$

$$\boldsymbol{\Sigma} \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i \quad i = 1 \dots D \quad \boldsymbol{u}_i{}^T \boldsymbol{u}_j = I_{ij}$$

- Covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{U}^T \boldsymbol{\Lambda} \boldsymbol{U}$ is a real symmetric matrix
- Matrix $\boldsymbol{U}^T = [\boldsymbol{u}_i \dots \boldsymbol{u}_d]$, therefore

$$\boldsymbol{\Sigma} = \sum_{i=1}^{D} \lambda_i \boldsymbol{u}_i \boldsymbol{u}_i{}^T \qquad \boldsymbol{\Sigma}^{-1} = \sum_{i=1}^{D} \frac{1}{\lambda_i} \boldsymbol{u}_i \boldsymbol{u}_i{}^T$$

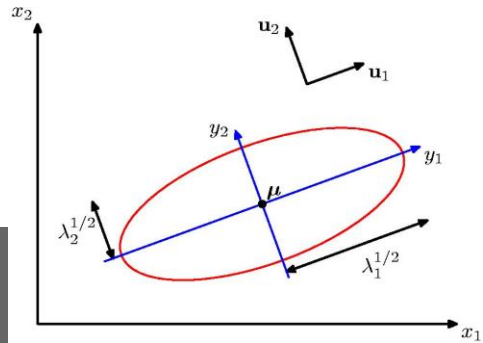
Gaussian Distribution

- Geometry of Multivariate Gaussian:

$$\Delta^2 = (\boldsymbol{x} - \boldsymbol{\mu})^T \sum_{i=1}^{D} \frac{1}{\lambda_i} \boldsymbol{u}_i \boldsymbol{u}_i{}^T (\boldsymbol{x} - \boldsymbol{\mu}) = \sum_{i=1}^{D} \frac{y_i{}^2}{\lambda_i}$$

- Change of coordinates:

$$y_i = \boldsymbol{u}_i{}^T (\boldsymbol{x} - \boldsymbol{\mu}) \quad \boldsymbol{y} = \boldsymbol{U}(\boldsymbol{x} - \boldsymbol{\mu})$$



Gaussian Distribution

- Geometry of Multivariate Gaussian: product of D univariate Gaussian distributions

$$J_{ij} = \frac{\partial x_i}{\partial y_j} = U_{ji} \qquad |\boldsymbol{J}| = 1 \qquad |\boldsymbol{\Sigma}|^{\frac{1}{2}} = \prod_{j=1}^{D} \lambda_j^{\frac{1}{2}}$$

$$p(\boldsymbol{y}) = p(\boldsymbol{x})|\boldsymbol{J}| = \prod_{j=1}^{D} \frac{1}{(2\pi\lambda_j)^{\frac{1}{2}}} \exp\left\{-\frac{{y_j}^2}{2\lambda_j}\right\}$$

$$\int p(\boldsymbol{y})d\boldsymbol{y} = \prod_{j=1}^{D} 1 = 1$$

Gaussian Distribution

- Moments: Expectation

$$\mathbb{E}[\boldsymbol{x}] = \frac{1}{(2\pi)^{\frac{D}{2}}} \frac{1}{|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \int \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\} \boldsymbol{x} d\boldsymbol{x}$$

$$= \frac{1}{(2\pi)^{\frac{D}{2}}} \frac{1}{|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \int \exp\left\{-\frac{1}{2}\boldsymbol{z}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{z}\right\} (\boldsymbol{z}+\boldsymbol{\mu}) d\boldsymbol{z}$$

$$= \frac{1}{(2\pi)^{\frac{D}{2}}} \frac{1}{|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \boldsymbol{\mu} \int \exp\left\{-\frac{1}{2}\boldsymbol{z}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{z}\right\} d\boldsymbol{z} = \boldsymbol{\mu}$$

Gaussian Distribution

- Geometry of Multivariate Gaussian: Covariance

$$\boldsymbol{z} = \boldsymbol{x} - \boldsymbol{\mu} \Rightarrow \boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

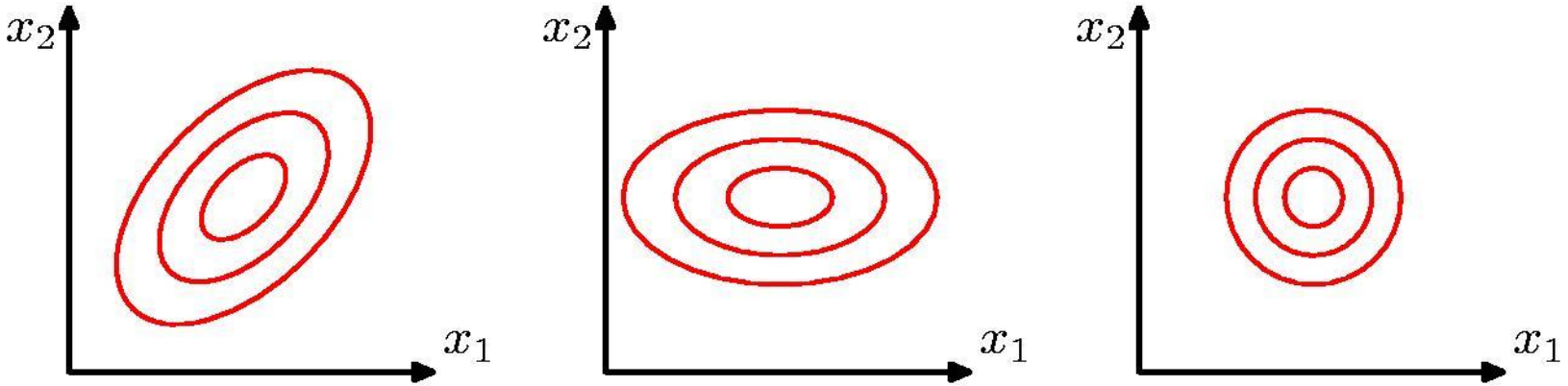
$$\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^T] = \mathbb{E}[(\boldsymbol{z} + \boldsymbol{\mu})(\boldsymbol{z} + \boldsymbol{\mu})^T]$$
$$= \mathbb{E}[\boldsymbol{\mu}\boldsymbol{\mu}^T + \boldsymbol{z}\boldsymbol{z}^T + \boldsymbol{\mu}\boldsymbol{z}^T + \boldsymbol{z}\boldsymbol{\mu}^T]$$
$$= \mathbb{E}[\boldsymbol{\mu}\boldsymbol{\mu}^T] + \mathbb{E}[\boldsymbol{z}\boldsymbol{z}^T] + \mathbb{E}[\boldsymbol{\mu}\boldsymbol{z}^T] + \mathbb{E}[\boldsymbol{z}\boldsymbol{\mu}^T] = \boldsymbol{\mu}\boldsymbol{\mu}^T + \mathbb{E}[\boldsymbol{z}\boldsymbol{z}^T]$$

$$\mathbb{E}[\boldsymbol{z}\boldsymbol{z}^T] = \mathbb{E}[(\boldsymbol{z} - \mathbf{0})(\boldsymbol{z} - \mathbf{0})^T] = \mathbb{E}[(\boldsymbol{z} - \mathbb{E}[\boldsymbol{z}])(\boldsymbol{z} - \mathbb{E}[\boldsymbol{z}])^T]$$
$$= cov(\boldsymbol{z}) = \boldsymbol{\Sigma}$$

$$\mathbb{E}[\boldsymbol{x}\boldsymbol{x}^T] = \boldsymbol{\mu}\boldsymbol{\mu}^T + \boldsymbol{\Sigma} \qquad cov(\boldsymbol{x}) = \mathbb{E}[\boldsymbol{x}\boldsymbol{x}^T] - \mathbb{E}[\boldsymbol{x}]\mathbb{E}[\boldsymbol{x}^T] = \boldsymbol{\Sigma}$$

Gaussian Distribution

- Shortcomings: $(D(D + 3)/2)$ parameters, unimodal
- Parameters reduction: $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_i^2)$ $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}$



Gaussian Distribution

- Conditional Gaussian Distributions
- If two variables are jointly Gaussian
  - Conditional distribution of one set on the other is also Gaussian
  - Marginal distribution is also Gaussian
- $\boldsymbol{x}$ is D-dimensional vector with $\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma})$ partitioned in two

$$\boldsymbol{x} = \begin{pmatrix} \boldsymbol{x}_a \\ \boldsymbol{x}_b \end{pmatrix} \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix} \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ab}{}^T & \boldsymbol{\Sigma}_{bb} \end{pmatrix} \quad \boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} \\ \boldsymbol{\Lambda}_{ab}{}^T & \boldsymbol{\Lambda}_{bb} \end{pmatrix}$$

$$\Delta^2 = (\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}-\boldsymbol{\mu})$$

$$= \left((\boldsymbol{x}_a-\boldsymbol{\mu}_a)^T (\boldsymbol{x}_b-\boldsymbol{\mu}_b)^T\right) \begin{pmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} \\ \boldsymbol{\Lambda}_{ab}{}^T & \boldsymbol{\Lambda}_{bb} \end{pmatrix} \begin{pmatrix} \boldsymbol{x}_a-\boldsymbol{\mu}_a \\ \boldsymbol{x}_b-\boldsymbol{\mu}_b \end{pmatrix}$$

Gaussian Distribution

- Conditional Gaussian Distributions: completing the square

$$
\begin{aligned}
&-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) \\
&= -\frac{1}{2}(\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x} - 2\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}) \\
&= -\frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x} + \boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + c
\end{aligned}
$$

Gaussian Distribution

- Conditional Gaussian Distributions: conditional covariance

$$\Delta^2 = (\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})$$
$$= (\boldsymbol{x}_a - \boldsymbol{\mu}_a)^T \boldsymbol{\Lambda}_{aa} (\boldsymbol{x}_a - \boldsymbol{\mu}_a) + (\boldsymbol{x}_a - \boldsymbol{\mu}_a)^T \boldsymbol{\Lambda}_{ab} (\boldsymbol{x}_b - \boldsymbol{\mu}_b)$$
$$+ (\boldsymbol{x}_b - \boldsymbol{\mu}_b)^T \boldsymbol{\Lambda}_{ab}{}^T (\boldsymbol{x}_a - \boldsymbol{\mu}_a) + (\boldsymbol{x}_b - \boldsymbol{\mu}_b)^T \boldsymbol{\Lambda}_{bb} (\boldsymbol{x}_b - \boldsymbol{\mu}_b)$$

- For $p(\boldsymbol{x}_a|\boldsymbol{x}_b)$ $\boldsymbol{x}_b$ is fixed: $p(\boldsymbol{x}_a|\boldsymbol{x}_b) \propto$
$$\exp\left\{-\frac{1}{2}\boldsymbol{x}_a{}^T \boldsymbol{\Lambda}_{aa} \boldsymbol{x}_a + \cdots\right\}$$

- For $\boldsymbol{x}_a$ the quadratic term "$\boldsymbol{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x}$" is $\boldsymbol{x}_a{}^T \boldsymbol{\Lambda}_{aa} \boldsymbol{x}_a$ is associated with matrix $\boldsymbol{\Lambda}_{aa} \Rightarrow \boldsymbol{\Sigma}_{a|b} = \boldsymbol{\Lambda}_{aa}{}^{-1}$

Gaussian Distribution

- Conditional Gaussian Distributions: conditional mean
- For $\boldsymbol{x}_a$ the linear term ${\boldsymbol{x}_a}^T {\boldsymbol{\Sigma}_{a|b}}^{-1} \boldsymbol{\mu}_{a|b}$ is associated with

$$\frac{1}{2}{\boldsymbol{x}_a}^T \boldsymbol{\Lambda}_{aa}\boldsymbol{\mu}_a + \frac{1}{2}{\boldsymbol{\mu}_a}^T \boldsymbol{\Lambda}_{aa}\boldsymbol{x}_a - \frac{1}{2}{\boldsymbol{x}_a}^T \boldsymbol{\Lambda}_{ab}(\boldsymbol{x}_b - \boldsymbol{\mu}_b) - \frac{1}{2}(\boldsymbol{x}_b - \boldsymbol{\mu}_b)^T {\boldsymbol{\Lambda}_{ab}}^T \boldsymbol{x}_a = {\boldsymbol{x}_a}^T \boldsymbol{\Lambda}_{aa}\boldsymbol{\mu}_a - {\boldsymbol{x}_a}^T \boldsymbol{\Lambda}_{ab}(\boldsymbol{x}_b - \boldsymbol{\mu}_b) = {\boldsymbol{x}_a}^T {\boldsymbol{\Sigma}_{a|b}}^{-1} \boldsymbol{\mu}_{a|b}$$

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\Sigma}_{a|b}\left(\boldsymbol{\Lambda}_{aa}\boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{ab}(\boldsymbol{x}_b - \boldsymbol{\mu}_b)\right)$$
$$= \boldsymbol{\mu}_a - {\boldsymbol{\Lambda}_{aa}}^{-1}\boldsymbol{\Lambda}_{ab}(\boldsymbol{x}_b - \boldsymbol{\mu}_b)$$

Gaussian Distribution

- Conditional Gaussian Distributions: Schur complement of $\boldsymbol{\Sigma}$ wrt. $\boldsymbol{D}$

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{M} & -\boldsymbol{M}\boldsymbol{B}\boldsymbol{D}^{-1} \\ \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{M} & \boldsymbol{D}^{-1} + \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{M}\boldsymbol{B}\boldsymbol{D}^{-1} \end{pmatrix}$$

$$\boldsymbol{M} = (\boldsymbol{A} - \boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{C})^{-1}$$

$$\begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ab}{}^{T} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} \\ \boldsymbol{\Lambda}_{ab}{}^{T} & \boldsymbol{\Lambda}_{bb} \end{pmatrix}$$

$$\boldsymbol{\Lambda}_{aa} = \left(\boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}{}^{-1}\boldsymbol{\Sigma}_{ab}{}^{T}\right)^{-1}$$

$$\boldsymbol{\Lambda}_{ab} = -\left(\boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}{}^{-1}\boldsymbol{\Sigma}_{ab}{}^{T}\right)^{-1}\boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}{}^{-1}$$

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a + \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}{}^{-1}(\boldsymbol{x}_b - \boldsymbol{\mu}_b)$$

$$\boldsymbol{\Sigma}_{a|b} = \boldsymbol{\Sigma}_{aa} - \boldsymbol{\Sigma}_{ab}\boldsymbol{\Sigma}_{bb}{}^{-1}\boldsymbol{\Sigma}_{ab}{}^{T}$$

Gaussian Distribution

- Marginal Gaussian Distributions:

$$p(\boldsymbol{x}_a) = \int p(\boldsymbol{x}_a, \boldsymbol{x}_b) d\boldsymbol{x}_b$$

- Considering $\boldsymbol{x}_b$ terms

$$-\frac{1}{2}\boldsymbol{x}_b{}^T\boldsymbol{\Lambda}_{bb}\boldsymbol{x}_b + \boldsymbol{x}_b{}^T\boldsymbol{\Lambda}_{bb}\boldsymbol{\mu}_b - \boldsymbol{x}_b{}^T\boldsymbol{\Lambda}_{ab}{}^T(\boldsymbol{x}_a - \boldsymbol{\mu}_a)$$

$$= -\frac{1}{2}\boldsymbol{x}_b{}^T\boldsymbol{\Lambda}_{bb}\boldsymbol{x}_b + \boldsymbol{x}_b{}^T\left(\boldsymbol{\Lambda}_{bb}\boldsymbol{\mu}_b - \boldsymbol{\Lambda}_{ab}{}^T(\boldsymbol{x}_a - \boldsymbol{\mu}_a)\right)$$

Gaussian Distribution

- Marginal Gaussian Distributions:

$$\boldsymbol{m} = \left(\boldsymbol{\Lambda}_{bb}\boldsymbol{\mu}_b - \boldsymbol{\Lambda}_{ab}{}^T(\boldsymbol{x}_a - \boldsymbol{\mu}_a)\right)$$

$$= -\frac{1}{2}\left(\boldsymbol{x}_b - \boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{m}\right)^T\boldsymbol{\Lambda}_{bb}\left(\boldsymbol{x}_b - \boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{m}\right)$$

$$+\frac{1}{2}\boldsymbol{m}^T\boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{m}$$

$$\int \exp\left\{-\frac{1}{2}\left(\boldsymbol{x}_b - \boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{m}\right)^T\boldsymbol{\Lambda}_{bb}\left(\boldsymbol{x}_b - \boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{m}\right)\right\}d\boldsymbol{x}_b$$

Gaussian Distribution

- Marginal Gaussian Distributions:

$$+\frac{1}{2}\left(\boldsymbol{\Lambda}_{bb}\boldsymbol{\mu}_b - \boldsymbol{\Lambda}_{ab}{}^T(\boldsymbol{x}_a - \boldsymbol{\mu}_a)\right)^T \boldsymbol{\Lambda}_{bb}{}^{-1}\left(\boldsymbol{\Lambda}_{bb}\boldsymbol{\mu}_b - \boldsymbol{\Lambda}_{ab}{}^T(\boldsymbol{x}_a - \boldsymbol{\mu}_a)\right) - \frac{1}{2}\boldsymbol{x}_a{}^T\boldsymbol{\Lambda}_{aa}\boldsymbol{x}_a + \boldsymbol{x}_a{}^T(\boldsymbol{\Lambda}_{aa}\boldsymbol{\mu}_a + \boldsymbol{\Lambda}_{ab}\boldsymbol{\mu}_b) + c_1$$

$$= -\frac{1}{2}\boldsymbol{x}_a{}^T\left(\boldsymbol{\Lambda}_{aa} - \boldsymbol{\Lambda}_{ab}\boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{\Lambda}_{ab}{}^T\right)\boldsymbol{x}_a + \boldsymbol{x}_a{}^T\left(\boldsymbol{\Lambda}_{aa} - \boldsymbol{\Lambda}_{ab}\boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{\Lambda}_{ab}{}^T\right)\boldsymbol{\mu}_a$$

Gaussian Distribution

- Marginal Gaussian Distributions:

$$\boldsymbol{\Sigma}_a = \left(\boldsymbol{\Lambda}_{aa} - \boldsymbol{\Lambda}_{ab}\boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{\Lambda}_{ab}{}^{T}\right)^{-1}$$

$$\boldsymbol{\Sigma}_a\left(\boldsymbol{\Lambda}_{aa} - \boldsymbol{\Lambda}_{ab}\boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{\Lambda}_{ab}{}^{T}\right)\boldsymbol{\mu}_a = \boldsymbol{\mu}_a$$

Schur complement:

$$\boldsymbol{\Sigma}_{aa} = \left(\boldsymbol{\Lambda}_{aa} - \boldsymbol{\Lambda}_{ab}\boldsymbol{\Lambda}_{bb}{}^{-1}\boldsymbol{\Lambda}_{ab}{}^{T}\right)^{-1}$$

$$\mathbb{E}[\boldsymbol{x}_a] = \boldsymbol{\mu}_a$$

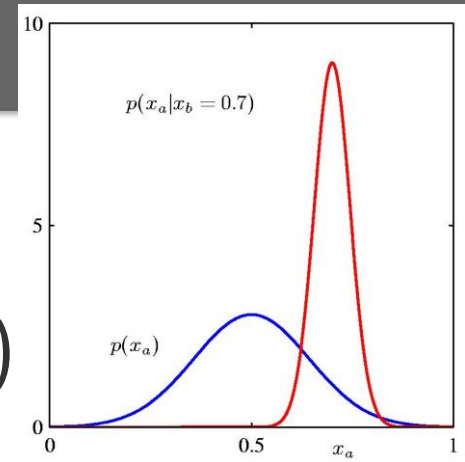
$$cov(\boldsymbol{x}_a) = \boldsymbol{\Sigma}_{aa}$$

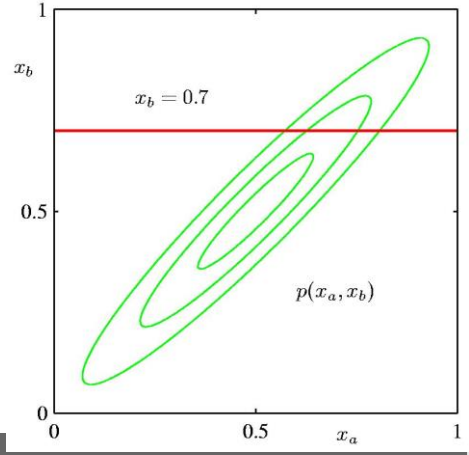
Gaussian Distribution

- Marginal Gaussian Distributions:

$$p(\boldsymbol{x}_a|\boldsymbol{x}_b) = \mathcal{N}\left(\boldsymbol{x}|(\boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}{}^{-1}\boldsymbol{\Lambda}_{ab}(\boldsymbol{x}_b - \boldsymbol{\mu}_b)), \boldsymbol{\Lambda}_{aa}{}^{-1}\right)$$

$$p(\boldsymbol{x}_a) = \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_{aa})$$





Gaussian Distribution

- Linear Gaussian Model:

$$p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$$
$$p(\boldsymbol{y}|\boldsymbol{x}) = \mathcal{N}(\boldsymbol{y}|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}, \boldsymbol{L}^{-1})$$

Mean is a linear function of $\boldsymbol{x}$ while covariance is independent of $\boldsymbol{x}$.

$$\boldsymbol{z} = \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix}$$

$$\ln p(\boldsymbol{z}) = \ln p(\boldsymbol{x}) + \ln p(\boldsymbol{y}|\boldsymbol{x})$$
$$= -\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Lambda}(\boldsymbol{x} - \boldsymbol{\mu}) - \frac{1}{2}(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b})^T \boldsymbol{L}(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) + c$$

Gaussian Distribution

- Bayes Theorem for Gaussian: covariance matrix

$$-\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{\Lambda}+\boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A})\boldsymbol{x}-\frac{1}{2}\boldsymbol{y}^T\boldsymbol{L}\boldsymbol{y}+\frac{1}{2}\boldsymbol{y}^T\boldsymbol{L}\boldsymbol{A}\boldsymbol{x}+\frac{1}{2}\boldsymbol{x}^T\boldsymbol{A}^T\boldsymbol{L}\boldsymbol{y}$$

$$=-\frac{1}{2}\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}^T\begin{pmatrix}\boldsymbol{\Lambda}+\boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A} & -\boldsymbol{A}^T\boldsymbol{L}\\ -\boldsymbol{L}\boldsymbol{A} & \boldsymbol{L}\end{pmatrix}\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}=-\frac{1}{2}\boldsymbol{z}^T\boldsymbol{R}\boldsymbol{z}$$

$$\boldsymbol{R}=\begin{pmatrix}\boldsymbol{\Lambda}+\boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A} & -\boldsymbol{A}^T\boldsymbol{L}\\ -\boldsymbol{L}\boldsymbol{A} & \boldsymbol{L}\end{pmatrix}$$

Gaussian Distribution

- Bayes Theorem for Gaussian: covariance matrix

$$\boldsymbol{R} = \begin{pmatrix} \boldsymbol{\Lambda} + \boldsymbol{A}^T \boldsymbol{L} \boldsymbol{A} & -\boldsymbol{A}^T \boldsymbol{L} \\ -\boldsymbol{L}\boldsymbol{A} & \boldsymbol{L} \end{pmatrix}$$

$$cov(\boldsymbol{z}) = \boldsymbol{R}^{-1} = \begin{pmatrix} \boldsymbol{\Lambda}^{-1} & \boldsymbol{\Lambda}^{-1}\boldsymbol{A}^T \\ \boldsymbol{A}\boldsymbol{\Lambda}^{-1} & \boldsymbol{L}^{-1} + \boldsymbol{A}\boldsymbol{\Lambda}^{-1}\boldsymbol{A}^T \end{pmatrix}$$

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{M} & -\boldsymbol{M}\boldsymbol{B}\boldsymbol{D}^{-1} \\ \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{M} & \boldsymbol{D}^{-1} + \boldsymbol{D}^{-1}\boldsymbol{C}\boldsymbol{M}\boldsymbol{B}\boldsymbol{D}^{-1} \end{pmatrix}$$

$$\boldsymbol{M} = (\boldsymbol{A} - \boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{C})^{-1}$$

Gaussian Distribution

- Bayes Theorem for Gaussian: mean
- Linear terms from exponent

$$\ln p(\boldsymbol{z}) = \ln p(\boldsymbol{x}) + \ln p(\boldsymbol{y}|\boldsymbol{x})$$

$$= -\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Lambda}(\boldsymbol{x}-\boldsymbol{\mu}) - \frac{1}{2}(\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b})^T\boldsymbol{L}(\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}) + c$$

$$\boldsymbol{x}^T\boldsymbol{\Lambda}\boldsymbol{\mu} - \boldsymbol{x}^T\boldsymbol{A}^T\boldsymbol{L}\boldsymbol{b} + \boldsymbol{y}^T\boldsymbol{L}\boldsymbol{b} = \begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{y}\end{pmatrix}^T \begin{pmatrix}\boldsymbol{\Lambda}\boldsymbol{\mu} - \boldsymbol{A}^T\boldsymbol{L}\boldsymbol{b}\\ \boldsymbol{L}\boldsymbol{b}\end{pmatrix}$$

$$\mathbb{E}[\boldsymbol{z}] = \boldsymbol{R}^{-1}\begin{pmatrix}\boldsymbol{\Lambda}\boldsymbol{\mu} - \boldsymbol{A}^T\boldsymbol{L}\boldsymbol{b}\\ \boldsymbol{L}\boldsymbol{b}\end{pmatrix} = \begin{pmatrix}\boldsymbol{\mu}\\ \boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b}\end{pmatrix}$$

Gaussian Distribution

- Bayes Theorem for Gaussian: finding $p(\boldsymbol{y})$

$$\mathbb{E}[\boldsymbol{z}] = \begin{pmatrix} \boldsymbol{\mu} \\ \boldsymbol{A\mu} + \boldsymbol{b} \end{pmatrix}$$

$$\mathrm{cov}(\boldsymbol{z}) = \begin{pmatrix} \boldsymbol{\Lambda}^{-1} & \boldsymbol{\Lambda}^{-1}\boldsymbol{A}^T \\ \boldsymbol{A\Lambda}^{-1} & \boldsymbol{L}^{-1} + \boldsymbol{A\Lambda}^{-1}\boldsymbol{A}^T \end{pmatrix}$$

$$\mathbb{E}[\boldsymbol{y}] = \boldsymbol{A\mu} + \boldsymbol{b}$$

$$\mathrm{cov}(\boldsymbol{y}) = \boldsymbol{L}^{-1} + \boldsymbol{A\Lambda}^{-1}\boldsymbol{A}^T$$

Gaussian Distribution

- Bayes Theorem for Gaussian: finding $p(\boldsymbol{x}|\boldsymbol{y})$

$$\boldsymbol{R} = \begin{pmatrix} \boldsymbol{\Lambda} + \boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A} & -\boldsymbol{A}^T\boldsymbol{L} \\ -\boldsymbol{L}\boldsymbol{A} & \boldsymbol{L} \end{pmatrix}$$

$$cov(\boldsymbol{x}|\boldsymbol{y}) = (\boldsymbol{\Lambda} + \boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A})^{-1}$$

- Linear terms wrt. $\boldsymbol{x}$:

$$-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Lambda}(\boldsymbol{x}-\boldsymbol{\mu}) - \frac{1}{2}(\boldsymbol{y}-\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x})^T\boldsymbol{L}(\boldsymbol{y}-\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x})$$

$$\boldsymbol{x}^T\boldsymbol{\Lambda}\boldsymbol{\mu} + \boldsymbol{x}^T\boldsymbol{A}^T\boldsymbol{L}(\boldsymbol{y}-\boldsymbol{b}) = \boldsymbol{x}^T(\boldsymbol{\Lambda}\boldsymbol{\mu} + \boldsymbol{A}^T\boldsymbol{L}(\boldsymbol{y}-\boldsymbol{b}))$$

$$\mathbb{E}[\boldsymbol{x}|\boldsymbol{y}] = (\boldsymbol{\Lambda} + \boldsymbol{A}^T\boldsymbol{L}\boldsymbol{A})^{-1}(\boldsymbol{\Lambda}\boldsymbol{\mu} + \boldsymbol{A}^T\boldsymbol{L}(\boldsymbol{y}-\boldsymbol{b}))$$

Gaussian Distribution

- Maximum Likelihood for Gaussian

$$\ln p(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = -\frac{ND}{2}\ln 2\pi - \frac{N}{2}\ln|\boldsymbol{\Sigma}| - \frac{1}{2}\sum_{n=1}^{N}(\boldsymbol{x}_n - \boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu})$$

$$= -\frac{1}{2}\left(\sum_{n=1}^{N}\boldsymbol{x}_n{}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x}_n - 2\sum_{n=1}^{N}\boldsymbol{x}_n{}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + c\right)$$

- Trace trick

$$\boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x} = tr(\boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x}) = tr(\boldsymbol{M}\boldsymbol{x}\boldsymbol{x}^T)$$

$$\sum_{n=1}^{N}\boldsymbol{x}_n{}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x}_n = tr\left(\boldsymbol{\Sigma}^{-1}\sum_{n=1}^{N}\boldsymbol{x}_n\boldsymbol{x}_n{}^T\right)$$

Gaussian Distribution

- Maximum Likelihood for Gaussian:

$$\frac{\partial(\ln p(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma}))}{\partial\boldsymbol{\mu}} = \frac{\partial}{\partial\boldsymbol{\mu}}\left(-\frac{1}{2}\sum_{n=1}^{N}(\boldsymbol{x}_n-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n-\boldsymbol{\mu})\right) =$$

- The partial derivative wrt. z_k picks terms where $i = j = k$:

$$f(\boldsymbol{z}) = \boldsymbol{z}^T\boldsymbol{M}\boldsymbol{z} = \sum_{ij} z_i M_{ij} z_j \qquad \boldsymbol{\nabla}\mathbf{z}(\boldsymbol{z}^T\boldsymbol{M}\boldsymbol{z}) = (\boldsymbol{M}+\boldsymbol{M}^T)\mathbf{z}$$

$$= -\frac{1}{2}\sum_{n=1}^{N} -2\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n-\boldsymbol{\mu}) = \sum_{n=1}^{N}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n-\boldsymbol{\mu}) \Rightarrow \boldsymbol{\mu}_{ML} = \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n$$

Gaussian Distribution

- Maximum Likelihood for Gaussian: covariance matrix

$$\boldsymbol{S} = \sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{\mu})(\boldsymbol{x}_n - \boldsymbol{\mu})^T$$

$$\frac{\partial}{\partial \boldsymbol{\Sigma}} (\ln p(\boldsymbol{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma})) = -\frac{N}{2} \frac{\partial}{\partial \boldsymbol{\Sigma}} (\ln|\boldsymbol{\Sigma}|) - \frac{1}{2} \frac{\partial}{\partial \boldsymbol{\Sigma}} [tr(\boldsymbol{\Sigma}^{-1}\boldsymbol{S})] = \frac{1}{2}(-N\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}^{-1}\boldsymbol{S}\boldsymbol{\Sigma}^{-1})$$

$$-N\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}^{-1}\boldsymbol{S}\boldsymbol{\Sigma}^{-1} = 0 \Rightarrow N\boldsymbol{\Sigma} = \boldsymbol{S} \Rightarrow \boldsymbol{\Sigma}_{ML} = \frac{1}{N} \sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{\mu}_{ML})(\boldsymbol{x}_n - \boldsymbol{\mu}_{ML})^T$$

Gaussian Distribution

- Maximum Likelihood for Gaussian: covariance matrix

$$\mathbb{E}[\boldsymbol{\mu}_{ML}] = \boldsymbol{\mu}$$

$$\mathbb{E}[\boldsymbol{\Sigma}_{ML}] = \frac{N-1}{N}\boldsymbol{\Sigma}$$

$$\boldsymbol{\Sigma}_{ML} = \frac{1}{N}\sum_{n=1}^{N}(\boldsymbol{x}_n - \boldsymbol{\mu}_{ML})(\boldsymbol{x}_n - \boldsymbol{\mu}_{ML})^T$$

$$\widetilde{\boldsymbol{\Sigma}} = \frac{1}{N-1}\sum_{n=1}^{N}(\boldsymbol{x}_n - \boldsymbol{\mu}_{ML})(\boldsymbol{x}_n - \boldsymbol{\mu}_{ML})^T$$

Gaussian Distribution

- *Sequential Estimation:*

$$\boldsymbol{\mu}_{ML}{}^{(N)} = \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n = \frac{1}{N}\boldsymbol{x}_N + \frac{1}{N}\sum_{n=1}^{N-1}\boldsymbol{x}_n$$

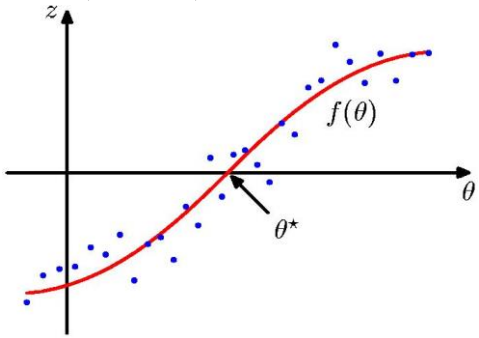
$$= \frac{1}{N}\boldsymbol{x}_N + \frac{N-1}{N}\boldsymbol{\mu}_{ML}{}^{(N-1)}$$

$$= \boldsymbol{\mu}_{ML}{}^{(N-1)} + \frac{1}{N}\left(\boldsymbol{x}_N - \boldsymbol{\mu}_{ML}{}^{(N-1)}\right)$$

Gaussian Distribution

- *Robbins-Monro algorithm*: z, θ joint distribution $p(z, \theta)$

$$f(\theta) \equiv \mathbb{E}[z|\theta] = \int zp(z|\theta)dz$$



- Seek θ^* such that $f(\theta^*) = 0$

$$\mathbb{E}[(z - f)^2|\theta] < \infty$$

$$\theta^{(N)} = \theta^{(N-1)} - a_{N-1} z(\theta^{(N-1)})$$

$$\lim_{n\to\infty} a_N = 0 \qquad \sum_{n=1}^{\infty} a_N = \infty \qquad \sum_{n=1}^{\infty} {a_N}^2 < \infty$$

Gaussian Distribution

- Solving MLE sequentially with Robbins-Monro algorithm:
- Algorithm is suitable as θ_{ML} is a stationary point
- Robbins-Monro algorithm finds the root of a function
- Instead of summing up gradients over all data at once, we do a small "step" in the direction of the instantaneous gradient from each new point

$$\frac{\partial}{\partial\theta}\left(-\frac{1}{N}\sum_{n=1}^{N}\ln p(x_n|\theta)\right)=0 \qquad -\lim_{n\to\infty}\frac{1}{N}\sum_{n=1}^{N}\frac{\partial}{\partial\theta}\ln p(x_n|\theta)=\mathbb{E}_x\left[-\frac{\partial}{\partial\theta}\ln p(x_n|\theta)\right]$$

Gaussian Distribution

- Solving MLE sequentially with Robbins-Monro algorithm:

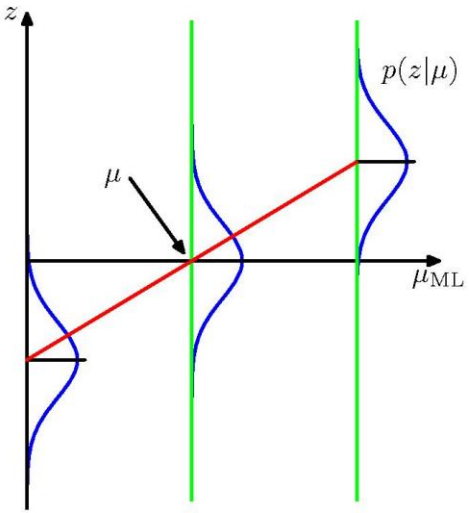
$$\theta^{(N)} = \theta^{(N-1)} - a_{N-1}\frac{\partial}{\partial\theta}\left(-\ln p\left(x_n\middle|\theta^{(N-1)}\right)\right)$$

- For ${\mu_{ML}}^{(N)}$

$$z = \frac{\partial}{\partial\mu_{ML}}\left(-\ln p(x|\mu_{ML},\sigma^2)\right) = \frac{\partial}{\partial\mu_{ML}}\left(\frac{(x-\mu_{ML})^2}{2\sigma^2}\right)$$

$$= -\frac{x-\mu_{ML}}{\sigma^2} \qquad a_{N-1} = \frac{\sigma^2}{N}$$

$${\mu_{ML}}^{(N)} = {\mu_{ML}}^{(N-1)} + \frac{x - {\mu_{ML}}^{(N-1)}}{N}$$



Gaussian Distribution

- Bayesian Inference for Gaussian: $\boldsymbol{X} = \{x_1, \dots, x_N\}$

$$p(\boldsymbol{X}|\mu) = \prod_{n=1}^{N} p(x_n|\mu)$$

$$= \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2\right\}$$

$$p(\mu) = \mathcal{N}(\mu|\mu_0, \sigma_0^2) \qquad p(\mu|\boldsymbol{X}) \propto p(\boldsymbol{X}|\mu)p(\mu)$$

Gaussian Distribution

- Bayesian Inference for Gaussian:

$$\ln p(\mu|\boldsymbol{X}) \propto -\frac{(\mu_0-\mu)^2}{2\sigma_0^2} - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2$$

$$= -\frac{\mu^2}{2\sigma_0^2} + \frac{\mu\mu_0}{\sigma_0^2} - \frac{{\mu_0}^2}{2\sigma_0^2} - \frac{N\mu^2}{2\sigma^2} + \frac{\mu}{\sigma^2}\sum_{n=1}^{N}x_n - \frac{1}{2\sigma^2}\sum_{n=1}^{N}{x_n}^2$$

$$= -\frac{1}{2}\left[\frac{1}{\sigma_0^2}+\frac{N}{\sigma^2}\right]\mu^2 + \left[\frac{\mu_0}{\sigma_0^2}+\frac{1}{\sigma^2}\sum_{n=1}^{N}x_n\right]\mu - \frac{1}{2}\left[\frac{{\mu_0}^2}{\sigma_0^2}+\frac{1}{\sigma^2}\sum_{n=1}^{N}{x_n}^2\right]$$

Gaussian Distribution

- Bayesian Inference for Gaussian: completing the square wrt. μ

$$-\frac{1}{2}\left[\frac{1}{\sigma_0^2}+\frac{N}{\sigma^2}\right]\mu^2+\left[\frac{\mu_0}{\sigma_0^2}+\frac{1}{\sigma^2}\sum_{n=1}^{N}x_n\right]\mu-\frac{1}{2}\left[\frac{{\mu_0}^2}{\sigma_0^2}+\frac{1}{\sigma^2}\sum_{n=1}^{N}{x_n}^2\right]$$

$$-\frac{1}{2\sigma_N^2}(\mu-\mu_N)^2=-\frac{1}{2\sigma_N^2}\mu^2+\frac{\mu_N}{\sigma_N^2}\mu-\frac{1}{2\sigma_N^2}{\mu_N}^2\Rightarrow$$

$$\sigma_N^2=\left(\frac{1}{\sigma_0^2}+\frac{N}{\sigma^2}\right)^{-1}\qquad\left[\frac{\mu_0}{\sigma_0^2}+\frac{1}{\sigma^2}\sum_{n=1}^{N}x_n\right]=\frac{\mu_N}{\sigma_N^2}$$

Gaussian Distribution

- Bayesian Inference for Gaussian: completing the square wrt. μ

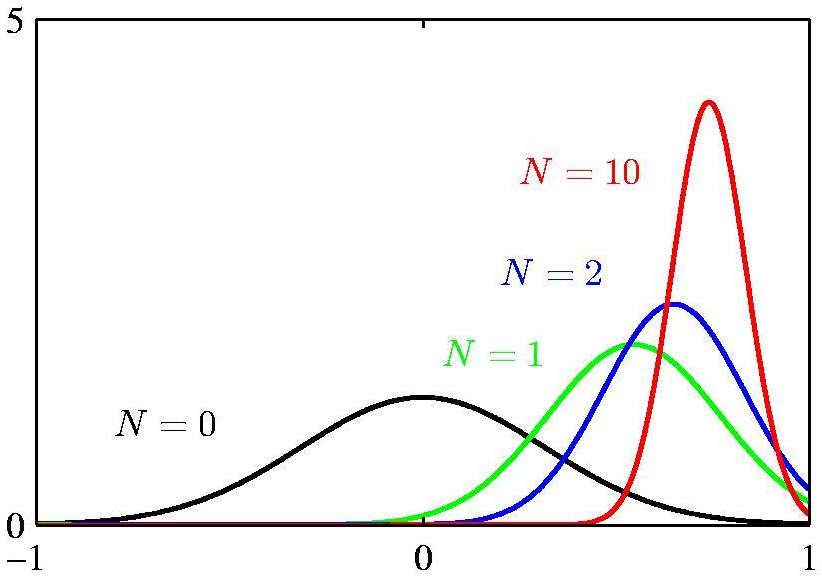
$$\mu_N = \left(\frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}\right)^{-1} \left[\frac{\mu_0}{\sigma_0^2} + \frac{1}{\sigma^2}\sum_{n=1}^{N} x_n\right]$$

$$= \frac{\sigma^2\sigma_0^2}{\sigma^2 + N\sigma_0^2}\left[\frac{\mu_0}{\sigma_0^2} + \frac{1}{\sigma^2}\sum_{n=1}^{N} x_n\right] = \frac{\sigma^2\mu_0}{\sigma^2 + N\sigma_0^2} + \frac{N\sigma_0^2}{\sigma^2 + N\sigma_0^2}\mu_{ML}$$

Gaussian Distribution

- Bayesian Inference for Gaussian:
- Compromise between μ_0 and μ_{ML}
- $N = 0$ leads to $\mu_N = \mu_0$
- $N \to \infty$ leads to $\mu_N = \mu_{ML}$
- Precision is additive and is steadily increasing
- $N = 0$ leads to $\sigma_N^2 = \sigma_0^2$
- $N \to \infty$ leads to $\sigma_N^2 \to 0$
- $N, \sigma_0^2 \to \infty, \sigma_N^2 = \sigma^2/N$

$$\sigma_N^2 = \left(\frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}\right)^{-1}$$



Gaussian Distribution

- Bayesian paradigm leads very naturally to a sequential view of the inference problem

$$p(\boldsymbol{\mu}|\mathcal{D}) \propto \left[p(\boldsymbol{\mu}) \prod_{n=1}^{N-1} p(\boldsymbol{x}_n|\boldsymbol{\mu}) \right] p(\boldsymbol{x}_N|\boldsymbol{\mu})$$

- [...] can be viewed as a prior distribution combined with the likelihood function associated with data point $\boldsymbol{x}_N$ to arrive at the posterior distribution after observing N data points

$$p(\boldsymbol{X}|\lambda) = \prod_{n=1}^{N} \mathcal{N}(x_n|\mu, \lambda^{-1}) \propto \lambda^{N/2} \exp\left\{ -\frac{\lambda}{2} \sum_{n=1}^{N} (x_n - \mu)^2 \right\}$$

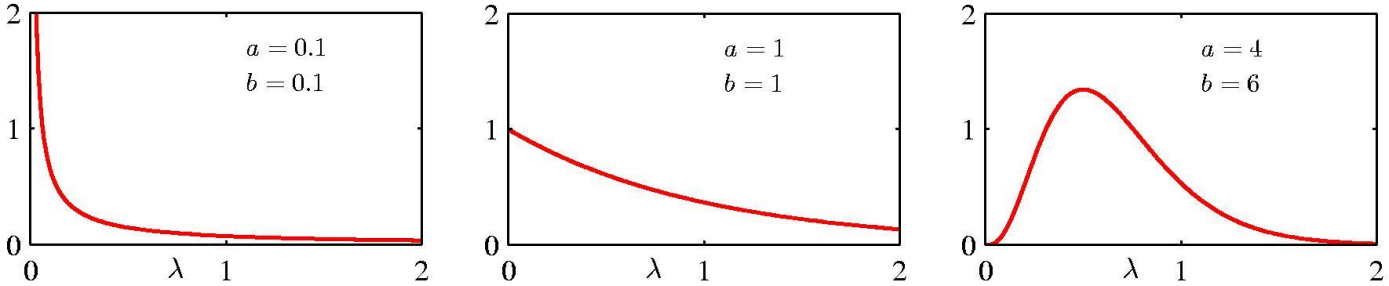
- Conjugate prior proportional to product of power of λ and exp of linear λ

Gaussian Distribution

- *Gamma distribution:*

$$Gam(\lambda|a,b) = \frac{1}{\Gamma(a)} b^a \lambda^{a-1} \exp(-b\lambda)$$

$$\Gamma(x) \equiv \int_0^\infty u^{x-1} e^{-u} du$$



Gaussian Distribution

- *Gamma distribution:*

$$\mathbb{E}[\lambda] = \frac{a}{b} \qquad var(\lambda) = \frac{a}{b^2}$$

$$p(X|\lambda)Gam(\lambda|a_0, b_0) \propto \lambda^{a_0-1}\lambda^{\frac{N}{2}} \exp\left\{-b_0\lambda - \frac{\lambda}{2}\sum_{n=1}^{N}(x_n-\mu)^2\right\}$$

$$= \lambda^{a_0+\frac{N}{2}-1} \exp\left\{-\lambda\left(b_0 + \frac{N}{2}\sigma_{ML}^2\right)\right\} = Gam\left(\lambda\middle|a_0 + \frac{N}{2}, b_0 + \frac{N}{2}\sigma_{ML}^2\right)$$

Gaussian Distribution

- *Gamma distribution:*

$$Gam\left(\lambda \middle| a_0 + \frac{N}{2}, b_0 + \frac{N}{2}\sigma_{ML}^2\right)$$

- Observing N data points increases a by $\frac{N}{2}$
- a_0 corresponds to $2a_0$ *effective* fictious prior observations
- b_0 is half the prior sum of squares
- $2b_0$ arising from $2a_0$ observations having variance $\frac{2b_0}{2a_0}$

Gaussian Distribution

- *Conjugate prior for unknown μ, λ:*

$$p(\boldsymbol{X}|\mu,\lambda) = \prod_{n=1}^{N} \left(\frac{\lambda}{2\pi}\right)^{1/2} \exp\left\{-\frac{\lambda}{2}(x_n-\mu)^2\right\}$$

$$\propto \left(\lambda^{1/2}\exp\left\{-\frac{\lambda\mu^2}{2}\right\}\right)^N \exp\left\{\lambda\mu\sum_{n=1}^{N} x_n - \frac{\lambda}{2}\sum_{n=1}^{N} {x_n}^2\right\}$$

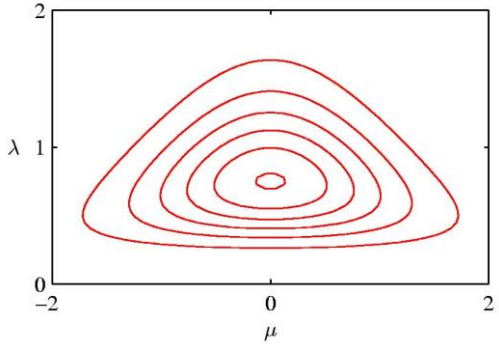
Gaussian Distribution

- *Conjugate prior for unknown μ, λ:*

$$p(\mu,\lambda) \propto \left(\lambda^{\frac{1}{2}}\exp\left\{-\frac{\lambda\mu^2}{2}\right\}\right)^{\beta} \exp\{c\lambda\mu - d\lambda\}$$

$$= \exp\left\{-\frac{\beta\lambda}{2}(\mu - c/\beta)^2\right\}\lambda^{\frac{\beta}{2}}\exp\left\{-\left(d - \frac{c^2}{2\beta}\right)\lambda\right\}$$

- Quadratic in μ.
- Linear in λ.
- Gamma distribution over λ.
- Independent of μ.



- *Gaussian Gamma distribution*

$$p(\mu,\lambda) = p(\mu|\lambda)p(\lambda) = \mathcal{N}\left(\mu\middle|\frac{c}{\beta},(\beta\lambda)^{-1}\right) Gam\left(\lambda\middle|\frac{\beta}{2}+1, d - \frac{c^2}{2\beta}\right)$$

Gaussian Distribution

- Multivariate case of D-dimensional variable $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$
- Conjugate prior for unknown $\boldsymbol{\mu}$ and known $\boldsymbol{\Lambda}$ is Gaussian
- Conjugate prior for known $\boldsymbol{\mu}$ and unknown $\boldsymbol{\Lambda}$ is Wishart
- Scale matrix $\boldsymbol{W}$, degrees of freedom ν, B normalization constant

$$\mathcal{W}(\boldsymbol{\Lambda}|\boldsymbol{W}, \nu) = B|\boldsymbol{\Lambda}|^{(\nu-D-1)/2} \exp\left\{-\frac{1}{2} tr(\boldsymbol{W}^{-1}\boldsymbol{\Lambda})\right\}$$

$$B(\boldsymbol{W}, \nu) = |\boldsymbol{W}|^{-\nu/2} \left(2^{\nu D/2} \pi^{D(D-1)/4} \prod_{i=1}^{D} \Gamma\left(\frac{\nu + 1 - i}{2}\right)\right)^{-1}$$

- Conjugate prior for unknown $\boldsymbol{\mu}$ and unknown $\boldsymbol{\Lambda}$ is Gaussian Wishart distribution

$$p(\boldsymbol{\mu}, \boldsymbol{\Lambda}|\boldsymbol{\mu}_0, \beta, \boldsymbol{W}, \nu) = \mathcal{N}(\boldsymbol{\mu}|\boldsymbol{\mu}_0, (\beta\boldsymbol{\Lambda})^{-1})\mathcal{W}(\boldsymbol{\Lambda}|\boldsymbol{W}, \nu)$$

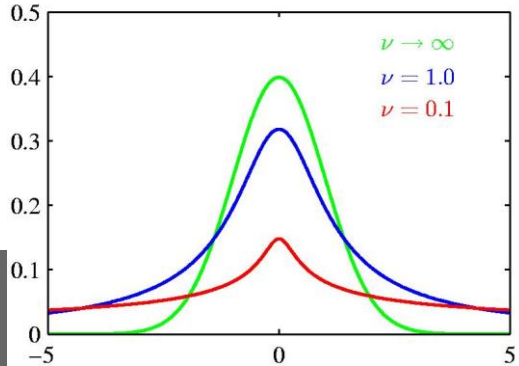
Student's t-distribution

- Marginal distribution
- $p(x|\mu,a,b) = \int_0^\infty \mathcal{N}(x|\mu,\tau^{-1})Gam(\tau|a,b)\,d\tau =$
$\int_0^\infty \left(\frac{\tau}{2\pi}\right)^{1/2} \exp\left\{-\frac{\tau}{2}(x_n-\mu)^2\right\}\frac{b^a \exp(-b\tau)\tau^{a-1}}{\Gamma(a)}d\tau =$
$\frac{b^a}{\Gamma(a)\sqrt{2\pi}}\int_0^\infty \exp\left\{-(b+\frac{1}{2}(x_n-\mu)^2)\tau\right\}\tau^{a-\frac{1}{2}}d\tau =$
- Standard result

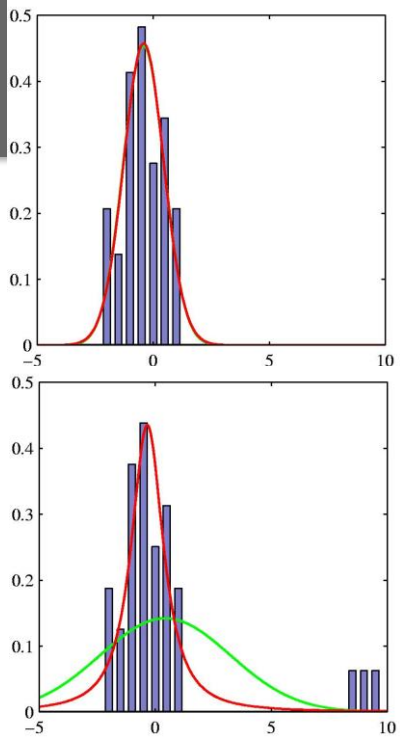
$$\int_0^\infty t^{\nu-1}\exp\{-\rho t\}dt = \frac{\Gamma(\nu)}{\rho^\nu} \quad t=\tau \quad \nu = a+\frac{1}{2} \quad \rho = b+\frac{1}{2}(x_n-\mu)^2$$

$$= \frac{b^a}{\Gamma(a)\sqrt{2\pi}}\frac{\Gamma\left(a+\frac{1}{2}\right)}{\left(b+\frac{1}{2}(x_n-\mu)^2\right)^{a+\frac{1}{2}}} = \left|\nu = 2a \;\; \lambda = \frac{a}{b}\right| =$$

$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma(\nu/2)}\left(\frac{\lambda}{\pi\nu}\right)^{1/2}\left(1+\frac{\lambda}{\nu}(x_n-\mu)^2\right)^{-\frac{\nu+1}{2}} = St(x|\mu,\lambda,\nu)$$



Student's t-distribution



- $St(x|\mu, \lambda, \nu)$, λ precision, ν degrees of freedom
- $\nu = 1$ Cauchy distribution
- $\nu \to \infty$ Gaussian $\mathcal{N}(x|\mu, \lambda^{-1})$
- Adding infinite amount of Gaussian distributions (Gaussian Mixture) with the same mean and different precision
- Student's t-distribution is *robust* and is relatively insensitive to *outliers*

Student's t-distribution

- Reformulation using $\nu = 2a \;\; \lambda = \frac{a}{b} \quad \eta = \tau \frac{b}{a}$

$$St(x|\mu,\lambda,\nu) = \int_0^\infty \mathcal{N}(x|\mu,(\eta\lambda)^{-1})Gam(\eta|\nu/2,\nu/2)\, d\eta$$

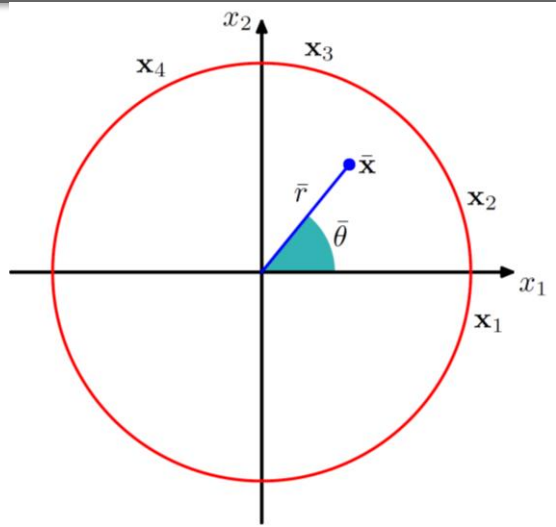
$$St(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Lambda},\nu) = \int_0^\infty \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},(\eta\boldsymbol{\Lambda})^{-1})Gam(\eta|\nu/2,\nu/2)\, d\eta$$

$$St(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Lambda},\nu) = \frac{\Gamma\left(\frac{\nu+D}{2}\right)}{\Gamma(\nu/2)}\left(\frac{|\boldsymbol{\Lambda}|}{(\pi\nu)^D}\right)^{1/2}(1+\frac{\Delta^2}{\nu})^{-\frac{\nu+D}{2}}$$

$$\Delta^2 = (\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Lambda}(\boldsymbol{x}-\boldsymbol{\mu}) \qquad \mathbb{E}[\boldsymbol{x}] = mode(\boldsymbol{x}) = \boldsymbol{\mu} \;\; cov(\boldsymbol{x}) = \frac{\nu}{\nu-2}\boldsymbol{\Lambda}^{-1}$$

Periodic variables

- Orientation, temporal cycles
- Discontinuity at the origin: $\theta_1 = 1°$, $\theta_2 = 359°$
  - Origin $0°$: $\mu = 180°$, $\sigma = 179°$
  - Origin $180°$: $\mu = 0°$, $\sigma = 1°$
- Observations $\mathcal{D} = \{\theta_1, \dots, \theta_N\}$ described by 2D unit vectors $\boldsymbol{x}_1, \dots, \boldsymbol{x}_N$ $\|\boldsymbol{x}_n\| = 1$



Periodic variables

$$\bar{\boldsymbol{x}} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n$$

$$\boldsymbol{x}_n = (\cos\theta_n, \sin\theta_n)$$

$$\bar{\boldsymbol{x}} = (\bar{r}\cos\bar{\theta}, \bar{r}\sin\bar{\theta})$$

$$= \left(\frac{1}{N}\sum_{n=1}^{N}\cos\theta_n, \frac{1}{N}\sum_{n=1}^{N}\sin\theta_n\right) \Rightarrow$$

$$\bar{\theta} = \tan^{-1}\frac{\sum_{n=1}^{N}\sin\theta_n}{\sum_{n=1}^{N}\cos\theta_n}$$

$$\boldsymbol{x}_n = (\cos\theta_n, \sin\theta_n)$$

$$\bar{\boldsymbol{x}} = (\bar{r}\cos\bar{\theta}, \bar{r}\sin\bar{\theta})$$

$$= \left(\frac{1}{N}\sum_{n=1}^{N}\cos\theta_n, \frac{1}{N}\sum_{n=1}^{N}\sin\theta_n\right) \Rightarrow$$

Periodic variables

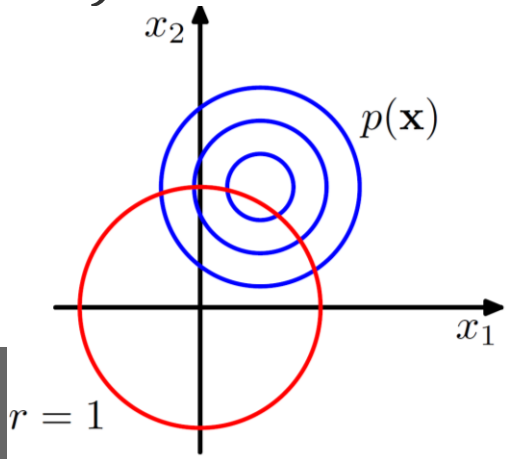
- *von Mises*: periodic generalization of the Gaussian

$$p(\theta) \geq 0 \quad \int_0^{2\pi} p(\theta)d\theta = 1 \quad p(\theta + M2\pi) = p(\theta)$$

$$p(x_1, x_2) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2}{2\sigma^2}\right\}$$

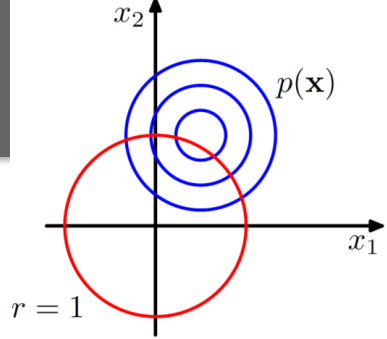
$$x_1 = r\cos\theta \qquad x_2 = r\sin\theta$$

$$\mu_1 = r_0\cos\theta_0 \qquad \mu_2 = r_0\sin\theta_0$$



Periodic variables

- *von Mises*: value distribution along the circle

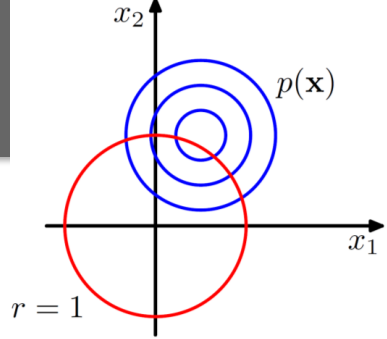


$$\exp\left\{-\frac{(\cos\theta - r_0\cos\theta_0)^2 + (\sin\theta - r_0\sin\theta_0)^2}{2\sigma^2}\right\}$$

$$= \exp\left\{-\frac{1}{2\sigma^2}(1 + {r_0}^2 - 2r_0\cos\theta\cos\theta_0 - 2r_0\sin\theta\sin\theta_0)\right\} = \exp\left\{\frac{r_0}{\sigma^2}\cos(\theta - \theta_0) + c\right\}$$

$$\cos A\cos B + \sin A\sin B = cos(A - B)$$

Periodic variables



- *von Mises*: circular normal distribution
- Mean θ_0, concentration m

$$p\left(\theta \middle| \theta_0, m = \frac{r_0}{\sigma^2}\right) = \frac{1}{2\pi I_0(m)} \exp\{m \cos(\theta - \theta_0)\}$$

$$I_0(m) = \frac{1}{2\pi} \int_0^{2\pi} \exp\{m \cos(\theta)\} d\theta$$

Periodic variables

- *von Mises*: Maximum Likelihood for mean θ_0^{ML}

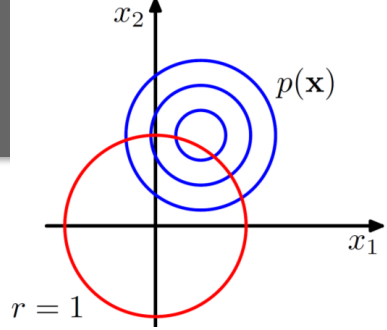
$$\ln p(\mathcal{D}|\theta_0, m) = -N \ln 2\pi - N I_0(m) + m \sum_{n=1}^{N} \cos(\theta_n - \theta_0)$$

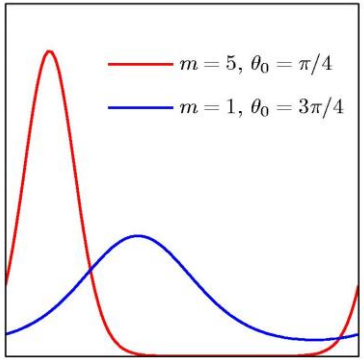
$$\frac{\partial}{\partial \theta_0} \ln p(\mathcal{D}|\theta_0, m) = \sum_{n=1}^{N} \sin(\theta_n - \theta_0) = 0$$

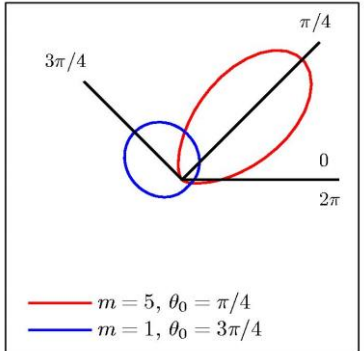
$$\sin(A - B) = \cos B \sin A - \cos A \sin B$$

$$\sum_{n=1}^{N} \sin(\theta - \theta_0) = 0 = \sum_{n=1}^{N} \cos\theta_0 \sin\theta_n - \cos\theta_n \sin\theta_0 \Rightarrow$$

$$\frac{\sin\theta_0}{\cos\theta_0} = \frac{\sum_{n=1}^{N} \sin\theta_n}{\sum_{n=1}^{N} \cos\theta_n} \Rightarrow \theta_0^{ML} = \tan^{-1} \frac{\sum_{n=1}^{N} \sin\theta_n}{\sum_{n=1}^{N} \cos\theta_n}$$







Periodic variables

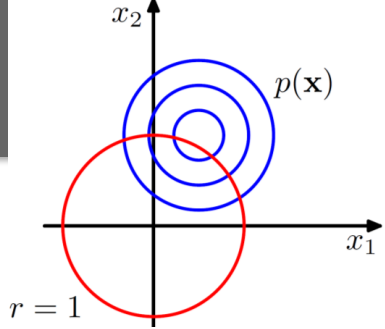
- *von Mises*: Maximum Likelihood for concentration m_{ML}

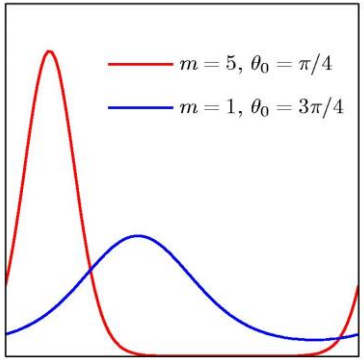
$$\ln p(\mathcal{D}|\theta_0, m) = -N \ln 2\pi - N I_0(m) + m \sum_{n=1}^{N} \cos(\theta_n - \theta_0)$$

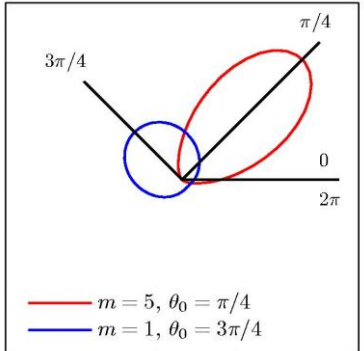
$$\frac{\partial}{\partial m} \ln p(\mathcal{D}|\theta_0, m) = \sum_{n=1}^{N} \cos(\theta_n - \theta_0^{ML}) - N \frac{I_1(m)}{I_0(m)}$$

$$A(m_{ML}) = \frac{1}{N} \sum_{n=1}^{N} \cos(\theta_n - \theta_0^{ML}) = \cos\theta_0^{ML} \frac{1}{N} \sum_{n=1}^{N} \cos\theta_n - \sin\theta_0^{ML} \frac{1}{N} \sum_{n=1}^{N} \sin\theta_n$$

$$= \frac{I_1(m_{ML})}{I_0(m_{ML})} = \frac{\int_0^{2\pi} \exp\{m_{ML}\cos(\theta)\}\cos(\theta)\, d\theta}{\int_0^{2\pi} \exp\{m_{ML}\cos(\theta)\} d\theta}$$





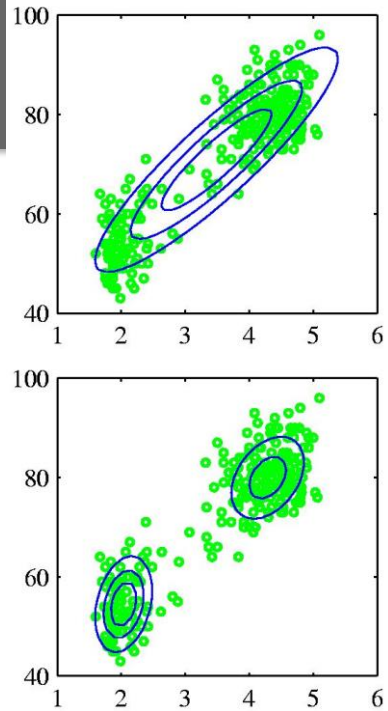


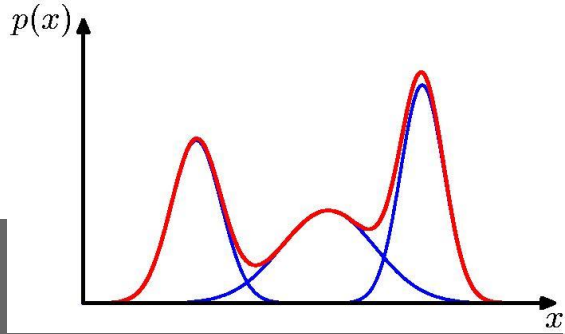
Mixtures of Gaussians

- Gaussian is unimodal
- Multimodal phenomena can be modeled using *mixture distributions*
- *Superposition of K Gaussian densities*

$$p(\boldsymbol{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = \sum_{k=1}^{K} p(k)p(\boldsymbol{x}|k)$$

$$1 \geq \pi_k \geq 0 \quad \sum_{k=1}^{K} \pi_k = 1$$



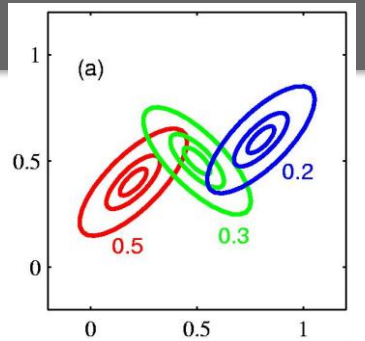


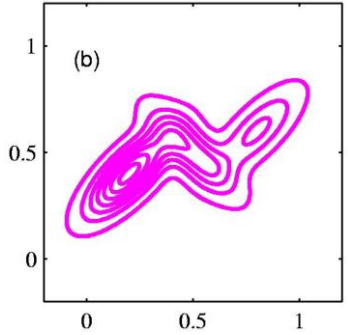
Mixtures of Gaussians

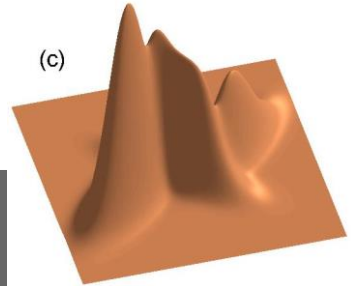
$$\gamma_k(\boldsymbol{x}) \equiv p(k|\boldsymbol{x}) = \frac{p(k)p(\boldsymbol{x}|k)}{\sum_l p(l)p(\boldsymbol{x}|l)} = \frac{\pi_k \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_l \pi_l \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)}$$

$$\ln p(\boldsymbol{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

- Sum inside the logarithm no longer closed-form analytical solution







Exponential Family

- General structure:

$$p(\boldsymbol{x}|\boldsymbol{\eta}) = h(\boldsymbol{x})g(\boldsymbol{\eta})\exp\{\boldsymbol{\eta}^T\boldsymbol{u}(\boldsymbol{x})\}$$

- $\boldsymbol{x}$ is scalar or vector, discrete or continuous
- $\boldsymbol{\eta}$ are natural parameters, $\boldsymbol{u}(\boldsymbol{x})$ some function of $\boldsymbol{x}$
- To normalize the distribution

$$g(\boldsymbol{\eta})\int h(\boldsymbol{x})\exp\{\boldsymbol{\eta}^T\boldsymbol{u}(\boldsymbol{x})\}\,d\boldsymbol{x} = 1$$

Exponential Family

- Bernoulli distribution:

$$p(x|\mu) = Bern(x|\mu) = \mu^x (1-\mu)^{1-x}$$

$$= \exp\{x \ln \mu + (1-x)\ln(1-\mu)\} = (1-\mu)\exp\left\{x \ln\left(\frac{\mu}{1-\mu}\right)\right\}$$

$$\eta = \ln\left(\frac{\mu}{1-\mu}\right) \Rightarrow e^{\eta} = \frac{\mu}{1-\mu} \Rightarrow \mu = \frac{e^{\eta}}{1+e^{\eta}} = \frac{1}{1+e^{-\eta}} = \sigma(\eta)$$

- *Logistic sigmoid function* $\sigma(\eta)$

$$p(x|\eta) = \big(1-\sigma(\eta)\big)\exp(\eta x) = \sigma(-\eta)\exp(\eta x)$$

$$u(x) = x \qquad h(x) = 1 \qquad g(\eta) = \sigma(-\eta)$$

Exponential Family

- Multinomial distribution:

$$p(\boldsymbol{x}|\boldsymbol{\mu}) = \prod_{k=1}^{M} \mu_k^{x_k} = \exp\left\{\sum_{k=1}^{M} x_k \ln \mu_k\right\}$$

$$p(\boldsymbol{x}|\boldsymbol{\eta}) = \exp\{\boldsymbol{\eta}^T \boldsymbol{x}\}, \eta_k = \ln \mu_k, u(x) = x, h(x) = g(\eta) = 1$$

- Only $M-1$ independent η_k because $\sum_{k=1}^{M} \mu_k = 1$
- Constraints $0 \leq \mu_k \leq 1$ $\sum_{k=1}^{M-1} \mu_k \leq 1$

Exponential Family

- Multinomial distribution:

$$\exp\left\{\sum_{k=1}^{M} x_k \ln \mu_k\right\} = \exp\left\{\sum_{k=1}^{M-1} x_k \ln \mu_k + x_M \ln \mu_M\right\}$$

$$= \exp\left\{\sum_{k=1}^{M-1} x_k \ln \mu_k + \left(1 - \sum_{k=1}^{M-1} x_k\right) \ln\left(1 - \sum_{k=1}^{M-1} \mu_k\right)\right\} = \exp\left\{\sum_{k=1}^{M-1} x_k \ln \mu_k\right.$$

$$\left. + \ln\left(1 - \sum_{k=1}^{M-1} \mu_k\right) - \ln\left(1 - \sum_{k=1}^{M-1} \mu_k\right) \sum_{k=1}^{M-1} x_k\right\} = \exp\left\{\sum_{k=1}^{M-1} x_k \ln \frac{\mu_k}{1 - \sum_{j=1}^{M-1} \mu_j}\right.$$

$$\left. + \ln\left(1 - \sum_{k=1}^{M-1} \mu_k\right)\right\} \Rightarrow \eta_k = \ln \frac{\mu_k}{1 - \sum_{j=1}^{M-1} \mu_j}$$

Exponential Family

- Multinomial distribution: summing both sides over k

$$\eta_k = \ln \frac{\mu_k}{1 - \sum_{j=1}^{M-1} \mu_j} \Rightarrow \mu_k = \frac{\exp \eta_k}{1 + \sum_{j=1}^{M-1} \exp \eta_k}$$

- *Softmax function (normalized exponential)*

$$p(\boldsymbol{x}|\boldsymbol{\eta}) = \left(1 - \sum_{k=1}^{M-1} \frac{\exp \eta_k}{1 + \sum_{j=1}^{M-1} \exp \eta_k}\right) \exp\{\boldsymbol{\eta}^T \boldsymbol{x}\} =$$

$$p(\boldsymbol{x}|\boldsymbol{\eta}) = \left(\frac{1 + \sum_{j=1}^{M-1} \exp \eta_k - \sum_{k=1}^{M-1} \exp \eta_k}{1 + \sum_{j=1}^{M-1} \exp \eta_k}\right) \exp\{\boldsymbol{\eta}^T \boldsymbol{x}\} = \left(1 + \sum_{k=1}^{M-1} \exp \eta_k\right)^{-1} \exp\{\boldsymbol{\eta}^T \boldsymbol{x}\}$$

$$\boldsymbol{u}(\boldsymbol{x}) = \boldsymbol{x}, h(\boldsymbol{x}) = 1, g(\boldsymbol{\eta}) = \left(1 + \sum_{k=1}^{M-1} \exp \eta_k\right)^{-1}$$

Exponential Family

- Gaussian distribution:

$$p(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}x^2 + \frac{\mu}{\sigma^2}x - \frac{1}{2\sigma^2}\mu^2\right\}$$

$$\boldsymbol{\eta} = \begin{pmatrix} \frac{\mu}{\sigma^2} \\ -\frac{1}{2\sigma^2} \end{pmatrix}, \boldsymbol{u}(x) = \begin{pmatrix} x \\ x^2 \end{pmatrix}, h(x) = 2\pi^{-1/2}, g(\boldsymbol{\eta}) = (-2\eta_2)^{1/2}\exp\left(\frac{\eta_1^2}{4\eta_2}\right)$$

Exponential Family

- Maximum Likelihood and Sufficient Statistics: estimating $\boldsymbol{\eta}$

$$\frac{\partial}{\partial \boldsymbol{\eta}}\left(g(\boldsymbol{\eta}) \int h(\boldsymbol{x}) \exp\{\boldsymbol{\eta}^T \boldsymbol{u}(\boldsymbol{x})\}\, d\boldsymbol{x} \right)$$

$$= \nabla g(\boldsymbol{\eta}) \int h(\boldsymbol{x}) \exp\{\boldsymbol{\eta}^T \boldsymbol{u}(\boldsymbol{x})\}\, d\boldsymbol{x} + g(\boldsymbol{\eta}) \int h(\boldsymbol{x}) \exp\{\boldsymbol{\eta}^T \boldsymbol{u}(\boldsymbol{x})\}\, \boldsymbol{u}(\boldsymbol{x})\, d\boldsymbol{x} \Rightarrow$$

$$-\nabla g(\boldsymbol{\eta}) \frac{1}{g(\boldsymbol{\eta})} = \mathbb{E}[\boldsymbol{u}(\boldsymbol{x})] = -\nabla \ln g(\boldsymbol{\eta})$$

- If we normalize the distribution from exponential family we find its moments by differentiation

$$\nabla^2(-\ln g(\boldsymbol{\eta})) = \nabla(-\nabla \ln g(\boldsymbol{\eta})) = \nabla \mathbb{E}[\boldsymbol{u}(\boldsymbol{x})] = \mathbb{E}[\boldsymbol{u}(\boldsymbol{x})\boldsymbol{u}(\boldsymbol{x})^T] - \mathbb{E}[\boldsymbol{u}(\boldsymbol{x})]\mathbb{E}[\boldsymbol{u}(\boldsymbol{x})]^T$$
$$= cov(\boldsymbol{u}(\boldsymbol{x}))$$

Exponential Family

- Maximum Likelihood and Sufficient Statistics $\boldsymbol{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_N\}$

$$p(\boldsymbol{X}|\boldsymbol{\eta}) = \left(\prod_{n=1}^{N} h(\boldsymbol{x}_n)\right) g(\boldsymbol{\eta})^N \exp\left\{\boldsymbol{\eta}^T \sum_{n=1}^{N} \boldsymbol{u}(\boldsymbol{x}_n)\right\}$$

$$\nabla_{\boldsymbol{\eta}} \ln p(\boldsymbol{X}|\boldsymbol{\eta}) = \nabla_{\boldsymbol{\eta}}\left(\ln\left(\prod_{n=1}^{N} h(\boldsymbol{x}_n)\right) + N \ln g(\boldsymbol{\eta}) + \boldsymbol{\eta}^T \sum_{n=1}^{N} \boldsymbol{u}(\boldsymbol{x}_n)\right) = 0$$

$$N\nabla_{\boldsymbol{\eta}} \ln g(\boldsymbol{\eta}) + \sum_{n=1}^{N} \boldsymbol{u}(\boldsymbol{x}_n) \Rightarrow -\nabla \ln g(\boldsymbol{\eta}_{ML}) = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{u}(\boldsymbol{x}_n)$$

- MLE depends on data only through $\sum_{n=1}^{N} \boldsymbol{u}(\boldsymbol{x}_n)$ which is sufficient statistics of $\boldsymbol{x}$
- For Bernoulli $\boldsymbol{u}(x) = x$ keeping sum of $\{x_n\}$ is sufficient
- For Gaussian $\boldsymbol{u}(x) = (x, x^2)^T$ keeping sum of $\{x_n\}$ and $\{x_n^2\}$

Exponential Family

- Conjugate Priors: for given likelihood $p(\boldsymbol{x}|\boldsymbol{\eta})$ we need prior $p(\boldsymbol{\eta})$ that is conjugate to it so that posterior $p(\boldsymbol{\eta}|\boldsymbol{x})$ has the same form as prior.

$$p(\boldsymbol{\eta}|\boldsymbol{\chi},\nu) = f(\boldsymbol{\chi},\nu)g(\boldsymbol{\eta})^{\nu}\exp\{\nu\boldsymbol{\eta}^T\boldsymbol{\chi}\}$$

$$p(\boldsymbol{\eta}|\boldsymbol{X},\boldsymbol{\chi},\nu) \propto p(\boldsymbol{\eta}|\boldsymbol{\chi},\nu)p(\boldsymbol{X}|\boldsymbol{\eta})$$

$$= f(\boldsymbol{\chi},\nu)g(\boldsymbol{\eta})^{\nu}\exp\{\nu\boldsymbol{\eta}^T\boldsymbol{\chi}\}\left(\prod_{n=1}^{N} h(\boldsymbol{x}_n)\right)g(\boldsymbol{\eta})^{N}\exp\{\boldsymbol{\eta}^T\boldsymbol{u}(\boldsymbol{x}_n)\}$$

$$= f(\boldsymbol{\chi},\nu)\left(\prod_{n=1}^{N} h(\boldsymbol{x}_n)\right)g(\boldsymbol{\eta})^{\nu+N}\exp\left\{\boldsymbol{\eta}^T\left(\sum_{n=1}^{N}\boldsymbol{u}(\boldsymbol{x}_n)+\nu\boldsymbol{\chi}\right)\right\}$$

$f(\boldsymbol{\chi},\nu)$ is a normalization factor, ν effective number of pseudoobservations each of which has a value for sufficient statistic $\boldsymbol{u}(\boldsymbol{x})$ given by $\boldsymbol{\chi}$

Exponential Family

- Noninformative Priors: letting the data speak for themselves
- "as objective as possible" Bayesian analysis
  - Help define a proper posterior
  - Avoid pathological estimates
- For discrete cases uniform distribution might be a choice $p(\lambda) = \frac{1}{K}$
- $\lambda \in [a, b]$ real and bounded: $p(\lambda) = \frac{1}{b-a}$
- For continuous $p(\lambda) = c$ unbounded λ the distribution cannot be normalized, prior is improper
- If $\lambda = \eta^2$ $\quad p_\eta(\eta) = p_\lambda(\lambda)\left|\frac{d\lambda}{d\eta}\right| = p_\lambda(\eta^2)2\eta = c2\eta$
- While $p_\lambda(\lambda)$ is constant, $p_\eta(\eta)$ is not constant

Exponential Family

- Noninformative Priors: *Translation invariant priors*
- For density of form where μ is location parameter
- $p(x|\mu) = f(x-\mu) = f((x+c)-(\mu+c)) = f(\hat{x}-\hat{\mu}) = p(\hat{x}|\hat{\mu})$
- Density is independent on the choice of origin
- Prior should be $A \leq \mu \leq B$ to be shifted to $A - c \leq \mu \leq B - c$

$$\int_A^B p(\mu)\,d\mu = \int_{A-c}^{B-c} p(\mu)\,d\mu = \int_A^B p(\mu - c)\,d\mu \Rightarrow p(\mu) = c_\mu$$

$$p(\mu) = c_\mu \Rightarrow \mathcal{N}(\mu|\mu_0, \sigma_0^2 \to \infty)$$

- Prior vanishes $\mu_N = \frac{\sigma^2\mu_0}{\sigma^2+N\sigma_0^2} + \frac{N\sigma_0^2}{\sigma^2+N\sigma_0^2}\mu_{ML}, \sigma_N^2 = \left(\frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}\right)^{-1}$

Exponential Family

- Noninformative Priors: *Scale invariant priors*

$$p(x|\sigma) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right), \hat{x} = cx \wedge \hat{\sigma} = c\sigma \wedge \Rightarrow p(\hat{x}|\hat{\sigma}) = \frac{1}{\hat{\sigma}} f\left(\frac{\hat{x}}{\hat{\sigma}}\right)$$

- *Corresponding prior over σ*

$$\int_A^B p(\sigma)\, d\sigma = \int_{A/c}^{B/c} p(\sigma)\, d\sigma = \int_A^B p\left(\frac{1}{c}\sigma\right)\frac{1}{c}\, d\sigma \Rightarrow p(\sigma) = p\left(\frac{1}{c}\sigma\right)\frac{1}{c}$$

- The only function that remains the same shape for any scale factor $c \neq 0$ and for any A and B is $p(\sigma) \propto \frac{1}{\sigma}$

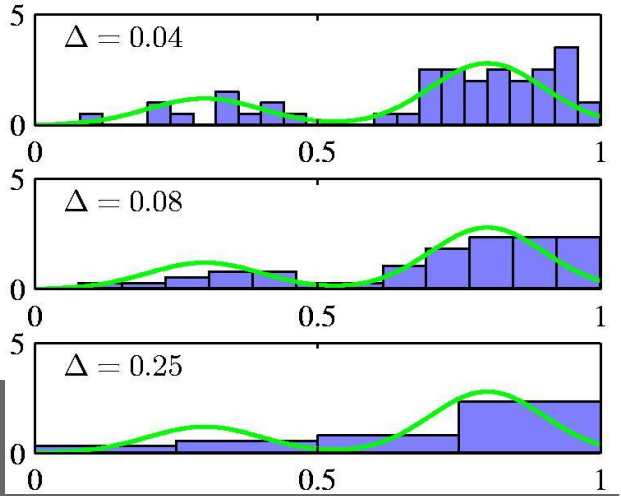
Exponential Family

- Noninformative Priors: *Scale invariant priors*
- Example is σ: $\mathcal{N}(x|\mu,\sigma^2) \propto \sigma^{-1}\exp\{-(x-\mu)/\sigma^2\}$
- If $\lambda = 1/\sigma^2$ and $p(\sigma) \propto 1/\sigma$ then $p(\lambda) \propto 1/\lambda$
- Working with λ prior can be $\mathrm{Gam}(\lambda|a_0,b_0)$
- Noniformative prior a special case of $a_0 = b_0 = 0$
- Posterior structure:

$$Gam\left(\lambda\middle|a_0+\frac{N}{2},b_0+\frac{N}{2}\sigma_{ML}^2\right) = \lambda^{a_0+\frac{N}{2}-1}\exp\left\{-\lambda\left(b_0+\frac{N}{2}\sigma_{ML}^2\right)\right\}$$

Nonparametric Methods

- *Histogram methods* for density estimation:
- Partitioning data into bins with width Δ_i (or Δ for const. size)
- Count number of observations n_i per bin
- $p_i = \frac{n_i}{N\Delta_i} \qquad \int p(x)dx = 1$
- Highly dependent on Δ and edge
- Can replace original data and supports sequential approach



Nonparametric Methods

- *Kernel Density Estimation*:
- Unknown probability density $p(\boldsymbol{x})$
- Probability mass: $P = \int_{\mathcal{R}} p(\boldsymbol{x}) d\boldsymbol{x}$
- For N observations from $p(\boldsymbol{x})$ each observation has probability P to fall into $\mathcal{R}$
- For K points falling inside $\mathcal{R}$ the probability follows binomial distribution $Bin(K|N,P) = \binom{N}{K} P^K \, (1-P)^{N-K}$
- $\mathbb{E}[{}^{K}/_{N}] = P, var[{}^{K}/_{N}] = P(1-P)/N$

Nonparametric Methods

- *Kernel Density Estimation:*
- $K \simeq NP$ for small region $\mathcal{R}$ with approx. constant $p(\boldsymbol{x})$
  $P \simeq p(\boldsymbol{x})V \Rightarrow p(\boldsymbol{x}) = \frac{K}{NV}$
- For volume V, $\mathcal{R}$ is a small hypercube centered at $\boldsymbol{x}$
- Kernel function Parzen window:

$$k(\boldsymbol{u}) = \begin{cases} 1, |u_i| \leq \frac{1}{2} \; i = 1 \ldots D \\ 0, \qquad otherwise \end{cases}$$

Nonparametric Methods

- *Kernel Density Estimation:*
- Sum of all points falling into region $\mathcal{R}$: $K = \sum_{n=1}^{N} k\left(\frac{\boldsymbol{x}-\boldsymbol{x}_n}{h}\right)$

$$p(\boldsymbol{x}) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{h^D}k\left(\frac{\boldsymbol{x}-\boldsymbol{x}_n}{h}\right)$$

- Reinterpret: Sum of N cubes centered on N data points $\boldsymbol{x}_n$
- Suffers from artificial discontinuities from hypercubes

Nonparametric Methods

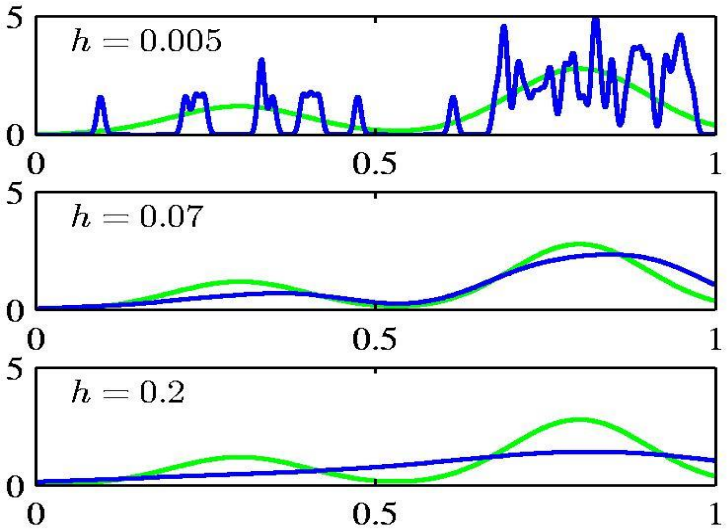
- *Kernel Density Estimation*:
- Smoother density model: Gaussian kernel

$$p(\boldsymbol{x}) = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{(2\pi h^2)^{D/2}}e^{-\frac{\|\boldsymbol{x}-\boldsymbol{x}_n\|^2}{2h^2}}$$

- Any kernel such that $k(\boldsymbol{u}) \geq 0$

$$\int k(\boldsymbol{u})d\boldsymbol{u} = 1$$

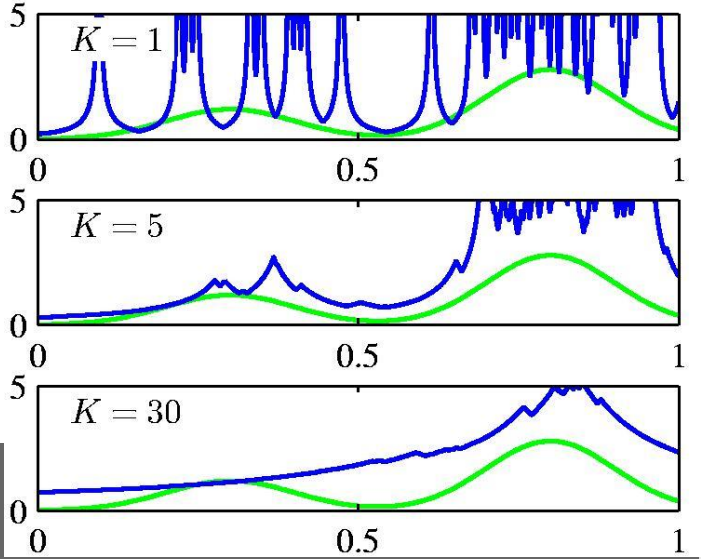
- No training phase, density evaluation scales with data size
- Issues with fixed h for irregular sampling



Nonparametric Methods

- *Nearest-Neighbor Methods*:
- Fixing K and finding suitable value for V
- Growing a sphere until it contains K data points
- Not a density model as integral over space diverges

$$p(\boldsymbol{x}) = \frac{K}{NV}$$



Nonparametric Methods

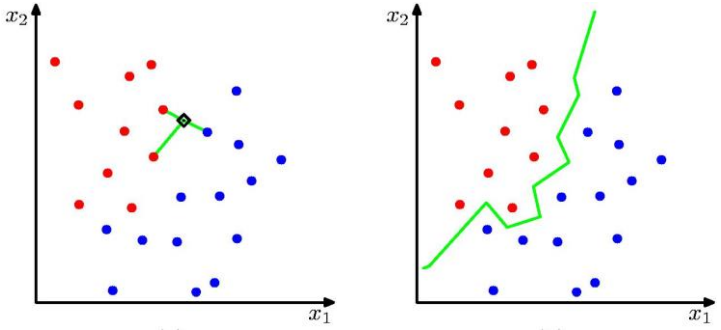
- *Nearest-Neighbor Methods*:
- Application of KNN density estimation to classification
- Total N points with N_k points in $\mathcal{C}_k$ class, $\sum N_k = N$
- Grow sphere to include K points of any class with volume V
- Containing K_k from $\mathcal{C}_k$ class

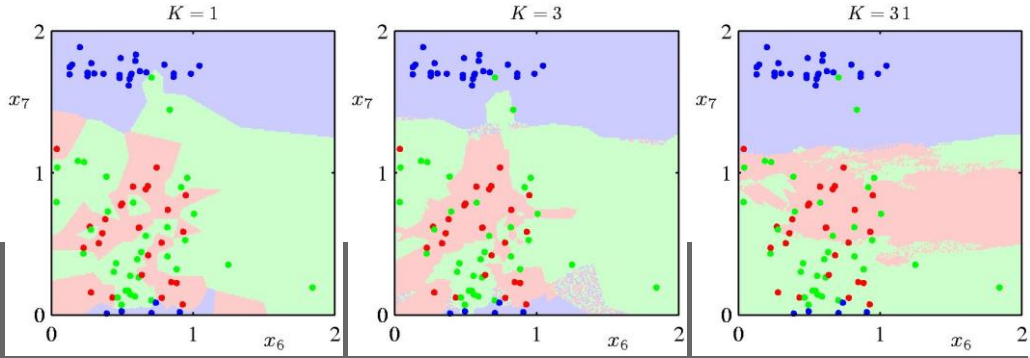
$$p(\boldsymbol{x}|\mathcal{C}_k) = \frac{K_k}{N_k V} \qquad p(\mathcal{C}_k) = \frac{N_k}{N} \qquad p(\boldsymbol{x}) = \frac{K}{NV}$$

$$p(\mathcal{C}_k|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\boldsymbol{x})} = \frac{K_k}{K}$$

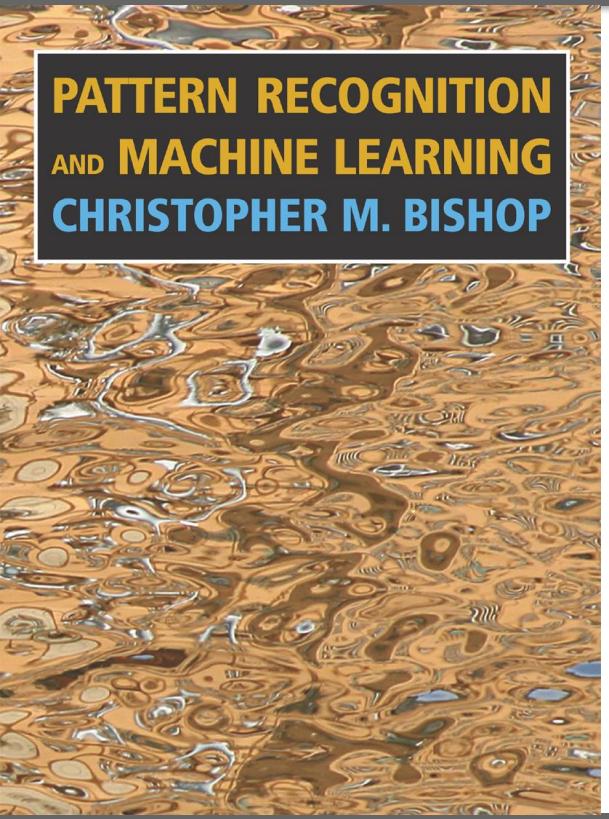
Nonparametric Methods

- *Nearest-Neighbor Methods*
- Minimize misclassification with choosing highest $p(\mathcal{C}_k|\boldsymbol{x})$
- K controls the degree of smoothing
- For faster evaluation uniform space partitioning





Credits



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